ITX4103/CSX4103 Requirement Engineering (541)

Term Project Final Report

“Pop Mart Auction Platform”

Team members

6530143 - Min Thuka

6530187 - Shwan Myat Nay Chi

6530201 - Lin Myat Phyo

6611201 - Aung Kaung Myat

6611040 - Sein Win Htut

ABSTRACT

This project aims to develop a secure, verification-based auction platform designed to minimize counterfeit sales and streamline the transaction process for collectible figures, particularly Pop Mart figures. The system comprises a web-based platform that allows sellers to list items for auction after administrative screening, while buyers can participate in real-time bidding with automatic auction closure. Additionally, a physical authenticity verification process and escrow payment mechanism are provided for administrators to ensure item legitimacy and secure financial handling, eliminating the risks of fraud and disputes associated with traditional peer-to-peer trust-based systems. This digital approach significantly reduces the potential for fake listings and payment defaults, ensuring that funds are only released after successful verification and delivery. The primary focus of the system is to enhance user confidence through authenticity assurance and reliable bidding, while simplifying the management of secure payments and logistics for the platform's administrative team.

Table of Contents

ABSTRACT *2*

PROJECT OVERVIEW *4*

Background **4**

Purpose **4**

Scope and Limitation **4**

REQUIREMENTS *5*

Business Model **5**

Requirement Gathering **5**

Stakeholders 5

Requirement Elicitation **6**

Understanding Buyer Behavior 6

Understanding Seller Workflows 7

Understanding Admin Responsibilities 8

Requirement Specification **9**

Buyers 9

Sellers 10

Admin 10

Functional Requirements 11

SYSTEM DESIGN *13*

Use Case Diagram **13**

Data Flow Diagram **14**

Entity Relation Diagram **24**

Data Dictionary **24**

USER INTERFACE *28*

UI/UX Design **28**

Prototype User Workflow **28**

CONCLUSION *41*

PROJECT OVERVIEW

Background

With the rapid growth of online marketplaces, auction-based platforms have become increasingly popular for buying and selling collectible items. However, many existing auction platforms face challenges related to counterfeit products, payment fraud, and disputes between buyers and sellers. These issues are particularly critical in collectible markets such as Pop Mart toys, where authenticity significantly impacts product value.

Traditional peer-to-peer auction systems often rely on trust between users, which can lead to scams, fake listings, or incomplete transactions. Buyers may hesitate to place high bids due to concerns about authenticity, while sellers may face delayed or failed payments. Additionally, the lack of a structured verification and escrow mechanism increases the risk of disputes and dissatisfaction.

To address these challenges, a secure, verification-based auction platform is required and our platform is the one that ensures safe bidding, reliable payment handling, and authenticity assurance for all transactions.

Purpose

The purpose of this project is to design and develop an auction platform that provides a secure, transparent, and trustworthy environment for trading Pop Mart collectible figures. The system aims to reduce fraud by introducing an escrow payment mechanism and a physical authenticity verification process managed by platform administrators.

The platform allows sellers to auction their items confidently, knowing that payments will be processed securely after verification. Buyers benefit from a reliable bidding system, real-time notifications, and assurance that purchased items are authentic before delivery. Overall, the system seeks to improve user confidence, reduce transaction disputes, and streamline auction operations through a well-defined digital workflow.

Scope and Limitation

The Pop Mart Auction Platform is a web-based system designed to support auction-based buying and selling of Pop Mart collectible items. The scope of the system includes:

1. Auction item listing by sellers after administrative screening
2. Real-time bidding with automatic auction closure
3. Escrow-based payment handling managed by administrators
4. Physical item collection and authenticity verification
5. Delivery of verified items to buyers
6. Seller payout after successful verification and delivery

However, the system has several limitations. The verification, logistics, and delivery process requires up to seven days and cannot be shortened due to operational constraints. The platform only supports Pop Mart-related items to ensure manageable and accurate authenticity checks. Additionally, sellers receive payment only after the buyer has received the item, which may not be suitable for users seeking immediate payouts.

REQUIREMENTS

Business Model

The business model of the Pop Mart Auction Platform follows an escrow-based auction workflow. Sellers submit items for auction listing, which are reviewed and approved by administrators. Once listed, buyers place bids in real time. When an auction ends, the highest bidder is required to make immediate payment, which is held in escrow by the platform.

The platform then arranges courier pickup of the item from the seller, conducts authenticity verification, and delivers the item to the buyer. After successful delivery, the platform deducts a service fee and releases the remaining payment to the seller. This model ensures trust, reduces fraud, and provides secure transactions for all parties involved.

Requirement Gathering

To identify system requirements accurately, requirement-gathering activities were conducted by analyzing stakeholder roles, workflows, and operational challenges. Stakeholders were categorized based on their interaction with the platform and their objectives within the system.

Stakeholders

1. Sellers
Sellers are users who list their Pop Mart items for auction. They are responsible for providing accurate item descriptions and shipping items for verification. Sellers expect fair auctions, secure payment handling, and timely payouts.

2.Buyers
Buyers are users who participate in auctions by placing bids on listed Pop Mart collectibles. They expect secure bidding, reliable notifications, authenticity assurance, and safe delivery of purchased items.

3. Admins
Admins manage platform operations, including listing approval, escrow payment holding, authenticity verification, dispute handling, and seller payouts. They ensure the overall integrity and reliability of the system.

Requirement Elicitation

To gather comprehensive requirements for the Pop Mart Auction Platform, multiple requirement elicitation techniques were employed, including interviews, online surveys, and workflow analysis. Each technique was specifically tailored to the roles, responsibilities, and concerns of different stakeholders, namely buyers, sellers, and platform administrators. This approach ensured that both functional needs and operational challenges were clearly identified.

Understanding Buyer Behavior

Online surveys were conducted to understand buyer preferences, bidding behavior, payment expectations, and trust-related concerns when participating in online auctions for collectible items.

Buyer Behavior Survey for Pop Mart Auction Platform

Section 1: Basic Information

1. What is your age group? (Multiple-choice question, required)
   - ☐ Below 18
   - ☐ 18–25
   - ☐ 26–35
   - ☐ Above 35
2. Have you purchased collectible items online before?
   - ☐ Yes
   - ☐ No

Section 2: Buyer Behavior

1. How often do you participate in online auctions?
   - ☐ Frequently
   - ☐ Occasionally
   - ☐ Rarely
   - ☐ Never
2. Have you ever faced issues when buying collectible items online?
   - ☐ Yes
   - ☐ No
   - ☐ Sometimes
3. If Yes or Sometimes, what issues did you face?
   - ☐ Fake or counterfeit items
   - ☐ Payment fraud
   - ☐ Delayed delivery
   - ☐ Seller not responding
   - ☐ Other: _____

4. What is your main concern when bidding on collectible items?
   - ☐ Item authenticity
   - ☐ Payment security
   - ☐ Delivery reliability
   - ☐ Auction fairness
5. Do you prefer real-time bidding notifications?
   - ☐ Yes, very important
   - ☐ Somewhat important
   - ☐ Not important
6. How would you like to receive bidding notifications?
   - ☐ Email
   - ☐ Mobile app notification
   - ☐ SMS
   - ☐ No notification needed
7. Would you feel more confident bidding if the platform verifies item authenticity before delivery?
   - ☐ Yes
   - ☐ No
   - ☐ Maybe
8. Would you continue using a platform that holds payment in escrow until delivery is completed?
   - ☐ Yes
   - ☐ No

Understanding Seller Workflows

Interviews and workflow observations were conducted with sellers to understand how they list items, manage auctions, and handle shipping and payments. These discussions focused on challenges related to trust, buyer behavior, and payout reliability.

Seller Workflow Understanding: Sample Interview Questions

Section 1: General Information

1. How long have you been selling collectible items online?
   - ☐ Less than 1 year
   - ☐ 1–3 years
   - ☐ 3–5 years
   - ☐ More than 5 years
2. What types of Pop Mart items do you usually sell?
   - ☐ Blind boxes
   - ☐ Limited editions

- ☐ Opened collectibles
- ☐ Sealed collectibles

Section 2: Auction Listing & Management

1. Can you describe your typical process for listing an item for auction? (Open-ended)
2. What challenges do you face when selling items through online auctions? (Open-ended)
3. Have you experienced cases where buyers failed to complete payment after winning an auction?
   - ☐ Yes, frequently
   - ☐ Sometimes
   - ☐ Rarely
   - ☐ Never
4. How do you currently ensure buyer trust in your listings?
   - ☐ Detailed photos
   - ☐ Accurate descriptions
   - ☐ Seller reviews
   - ☐ Platform verification
5. How important is fast payment after auction completion to you?
   - ☐ Extremely important
   - ☐ Somewhat important
   - ☐ Not important

Section 3: Shipping & Payment

1. How do you usually ship items to buyers?
   - ☐ Direct shipping to buyer
   - ☐ Through a third-party courier
   - ☐ Platform-arranged courier
2. Would you be willing to send items to a verification center before delivery?
   - ☐ Yes
   - ☐ No
   - ☐ Depends on payout guarantee
3. What is your preferred payout method?
   - ☐ Bank transfer
   - ☐ Digital wallet
   - ☐ Other: _____

Understanding Admin Responsibilities

Interviews were conducted with platform operators to understand administrative workflows, payment control, verification responsibilities, and dispute handling requirements.

Admin Responsibilities: Sample Interview Questions

Section 1: Auction & Platform Control

1. How do you screen and approve auction listings? (Open-ended)
2. How do you monitor ongoing auctions to prevent fraud or unfair bidding? (Open-ended)
3. What tools are needed to manage real-time auction activities effectively?

Section 2: Payment & Escrow Management

How should payments be handled after an auction ends?
- ☐ Immediate seller payout
- ☐ Hold payment in escrow
- ☐ Release payment after delivery

1. What challenges do you face in handling disputes between buyers and sellers? (Open-ended)
2. How long should escrow payments be held before seller payout?
   - ☐ Less than 3 days
   - ☐ 3–7 days
   - ☐ More than 7 days

Section 3: Verification & Reporting

1. What criteria are used to verify item authenticity? (Open-ended)
2. What actions should be taken if an item fails verification?
   - ☐ Refund buyer
   - ☐ Return item to seller
   - ☐ Suspend seller account
3. What type of reports are required for platform management?
   - ☐ Transaction reports
   - ☐ Revenue reports
   - ☐ Fraud and dispute logs

Requirement Specification

Based on the elicited requirements, stakeholder needs were documented using structured problem statements and translated into functional requirements.

Buyers

Problem ID	Problem Statement
prob_buy_1	A buyer wants to bid safely without fear of fraud
prob_buy_2	A buyer wants to be notified when outbid
prob_buy_3	A buyer wants assurance that items are authentic
prob_buy_4	A buyer wants secure payment processing after winning

prob_buy_5	A buyer wants to receive a refund if the item fails verification
prob_buy_6	A buyer wants to track delivery status after purchase
prob_buy_7	A buyer wants payment status visibility from the payment gateway
prob_buy_8	A buyer wants to save payment method and shipping address on their profile page for checkout simulation
prob_buy_9	A buyer wants clear handling if payment fails due to insufficient balance after winning an auction

Table 1: Problem Statement of Buyers

Sellers

Problem ID	Problem Statement
prob_sell_1	A seller wants secure payment after auction completion
prob_sell_2	A seller wants protection from non-paying buyers
prob_sell_3	A seller wants to set auction duration within allowed limits
prob_sell_4	A seller wants to be notified when their item is sold
prob_sell_5	A seller wants to track the verification status of their item
prob_sell_6	A seller wants transparency on service fee deductions
prob_sell_7	A seller wants a clear reason if their listing is rejected during admin screening
prob_sell_8	A seller wants to save payout method and address details on their profile page for settlement simulation

Table 2: Problem Statement of Sellers

Admin

Problem ID	Problem Statement
prob_admin_1	An admin wants to control escrow payments securely
prob_admin_2	An admin wants to verify the authenticity of submitted items
prob_admin_3	An admin wants to monitor live auctions to prevent suspicious activity
prob_admin_4	An admin wants to manage disputes between buyers and sellers
prob_admin_5	An admin wants the ability to suspend or ban fraudulent users
prob_admin_6	An admin wants to generate transaction and revenue reports
prob_admin_7	An admin wants to approve or reject seller listing requests before any auction goes live
prob_admin_8	An admin wants rejection decisions to include a mandatory reason

Table 3: Problem Statement of Admin

Functional Requirements

From Problem ID	Requirement ID	Functional Requirement
prob_buy_1	req_buy_1	The system shall allow buyers to place real-time bids.
prob_buy_2	req_buy_2	The system shall notify buyers when they are outbid.
prob_buy_3	req_buy_3	The system shall verify items before delivery.
prob_buy_4	req_buy_4	The system shall process payment automatically after an auction win.
prob_buy_5	req_buy_5	The system shall refund buyers if an item fails authenticity verification.
prob_buy_6	req_buy_6	The system shall provide shipment tracking information to buyers.
prob_buy_7	req_buy_7	The system shall show buyers payment gateway status updates for their winning auction payments.
prob_buy_8	req_buy_8	The system shall allow buyers to save and update payment method and shipping address details on the profile page.
prob_buy_9	req_buy_9	The system shall detect insufficient balance on winner payment, mark payment as failed, notify the buyer, and block auction completion until a successful payment retry within policy.
prob_sell_1	req_sell_1	The system shall release payment to sellers after verification and successful delivery.
prob_sell_2	req_sell_2	The system shall prevent auction completion if payment is not successful.
prob_sell_3	req_sell_3	The system shall allow sellers to select auction duration within defined limits.
prob_sell_4	req_sell_4	The system shall notify sellers when their item is sold.
prob_sell_5	req_sell_5	The system shall allow sellers to track item verification status.
prob_sell_6	req_sell_6	The system shall clearly display platform service fee deductions.
prob_sell_7	req_sell_7	The system shall show sellers the rejection reason when a listing request is rejected by admin.
prob_sell_8	req_sell_8	The system shall allow sellers to save and update payout method and address details on the profile page.
prob_admin_1	req_admin_1	The system shall hold buyer payments in escrow until verification is complete.

prob_admin_2	req_admin_2	The system shall manage the authenticity verification workflow.
prob_admin_3	req_admin_3	The system shall allow admins to monitor live bidding activities.
prob_admin_4	req_admin_4	The system shall allow admins to resolve disputes between users.
prob_admin_5	req_admin_5	The system shall allow admins to suspend or ban user accounts.
prob_admin_6	req_admin_6	The system shall generate transaction and revenue reports for platform analysis.
prob_admin_7	req_admin_7	The system shall require admin approval before a seller listing is published as a live auction.
prob_admin_8	requ_admin_8	The system shall require admins to enter a rejection reason when rejecting a seller listing request.

Table 4: Functional Requirements

SYSTEM DESIGN

Use Case Diagram

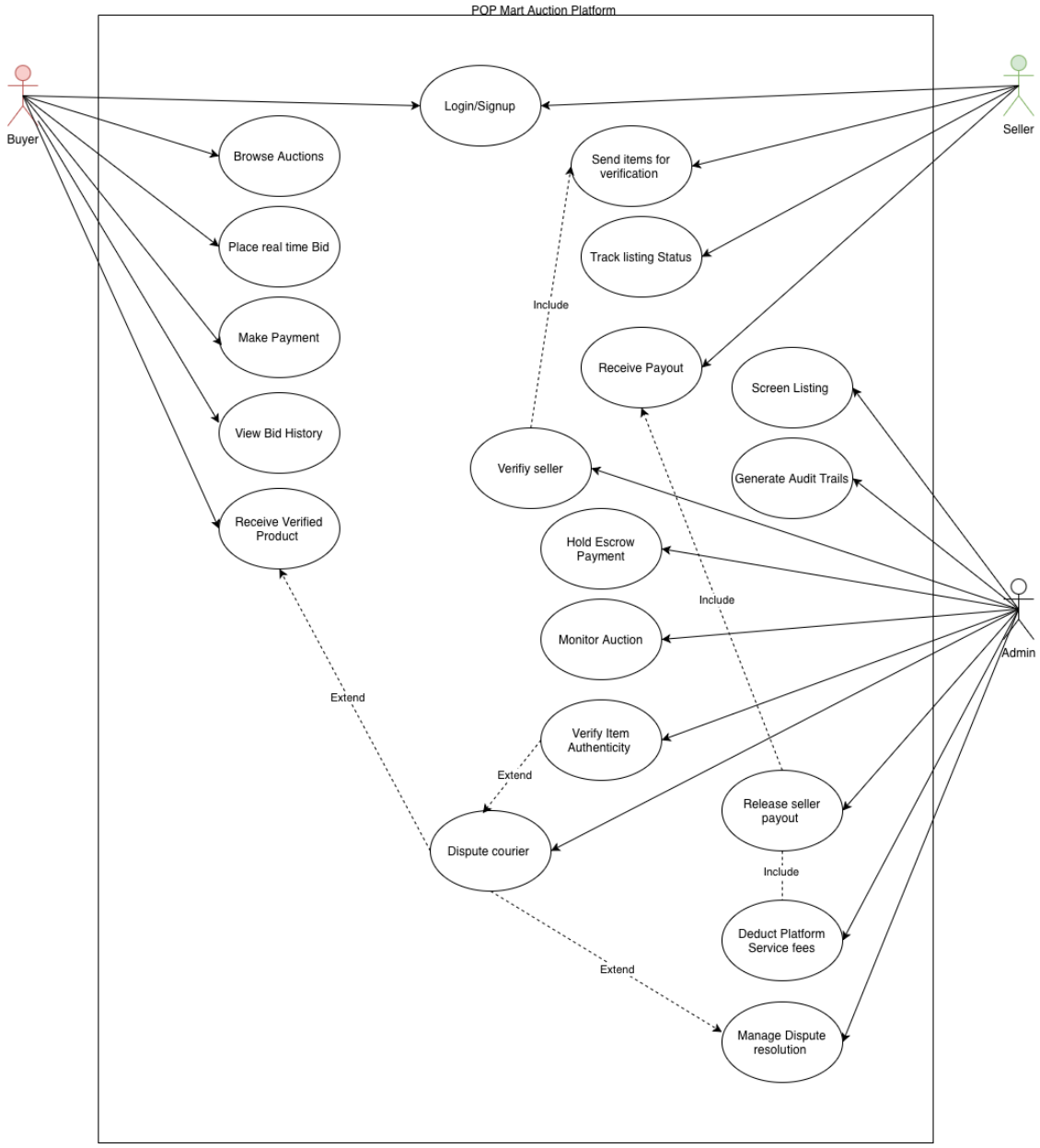

Figure 1: Use Case Diagram

Data Flow Diagram

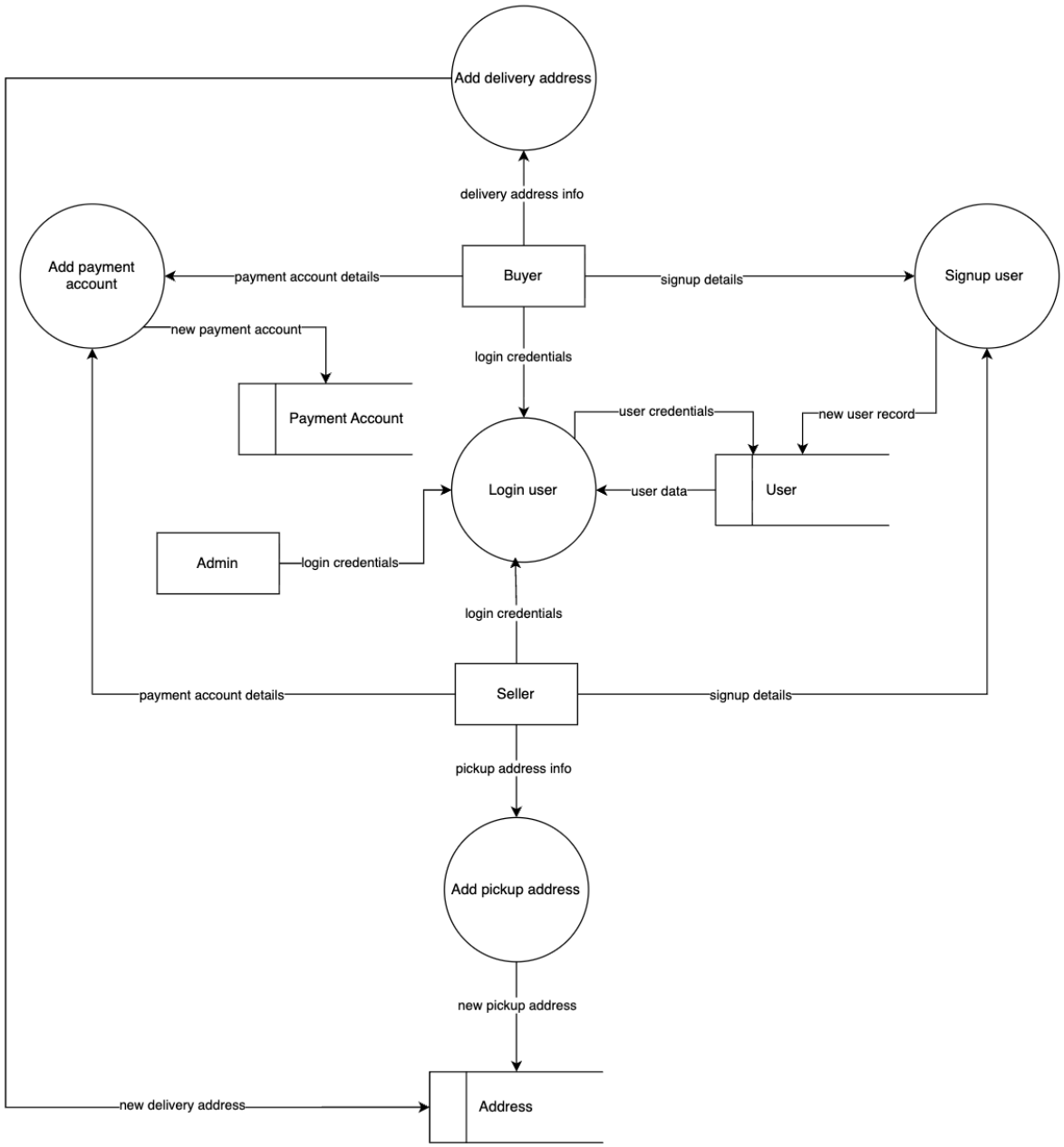

Figure 2: Manage User Account

Requirement ID: req_sell_8, req_buy_8

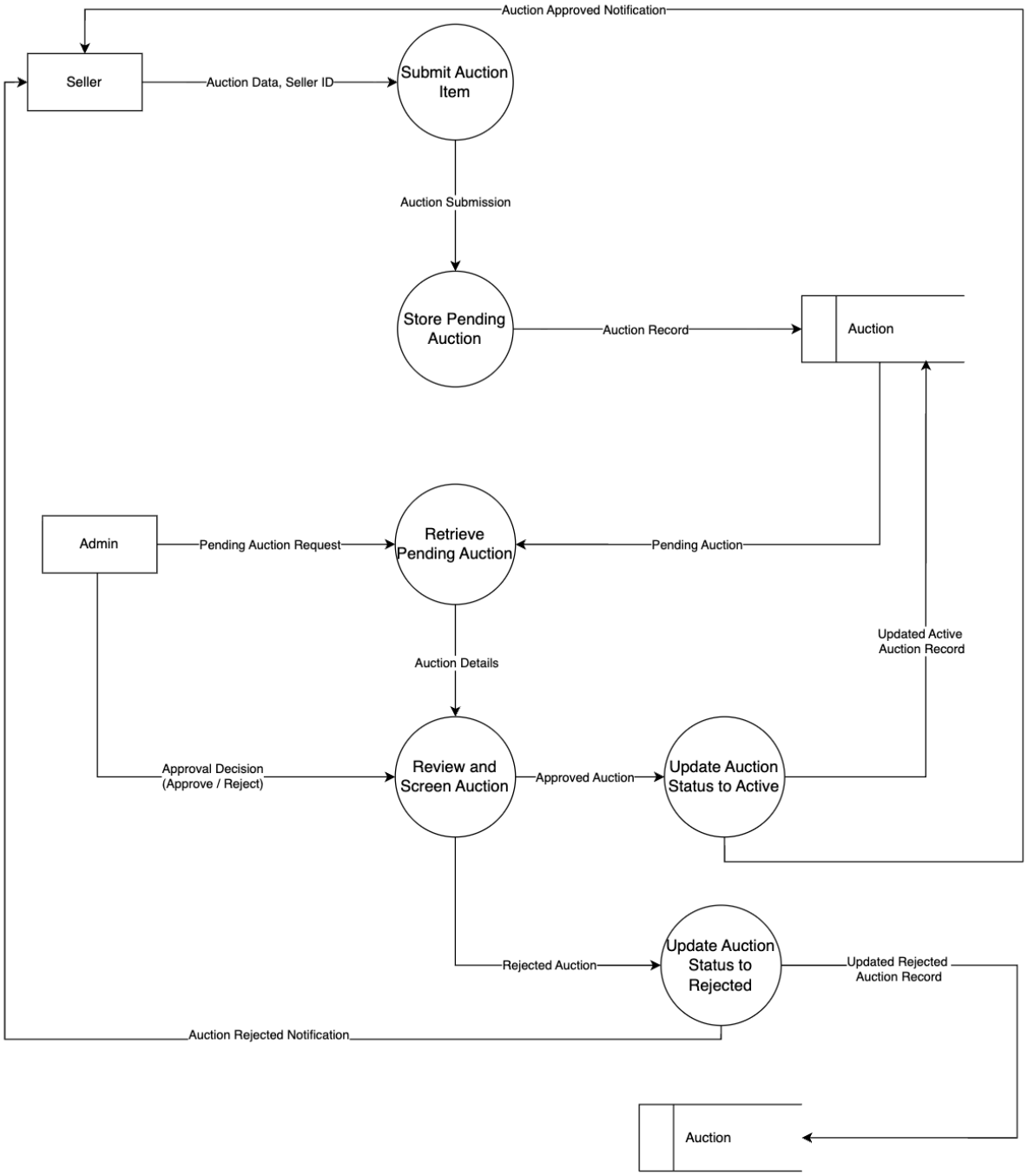

Figure 3: Submit Auction and Approval

Requirement ID: req_sell_3, req_admin_7, req_sell_7, req_admin_8

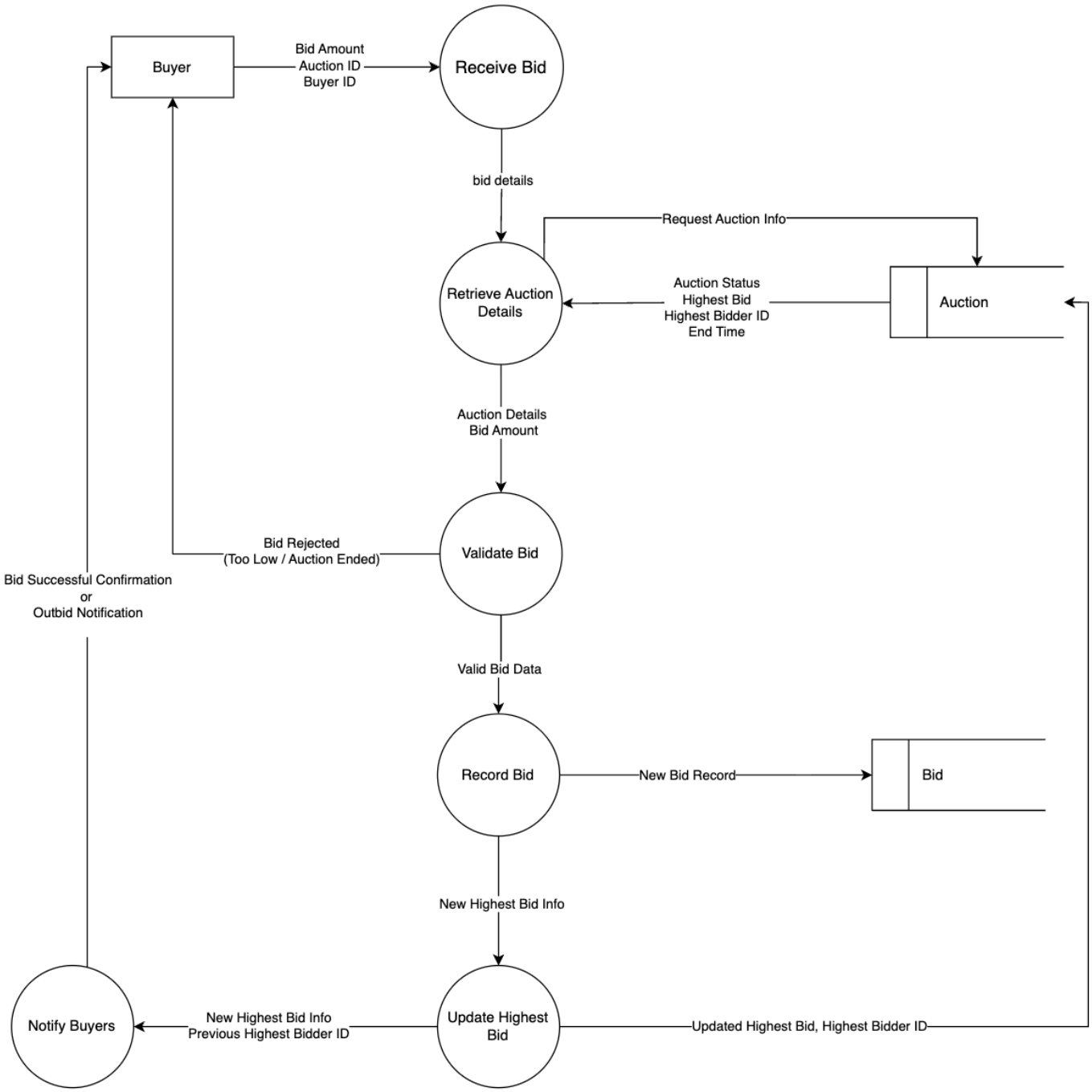

Figure 4: Manage Bidding

Requirement ID: req_buy_1, req_buy_2

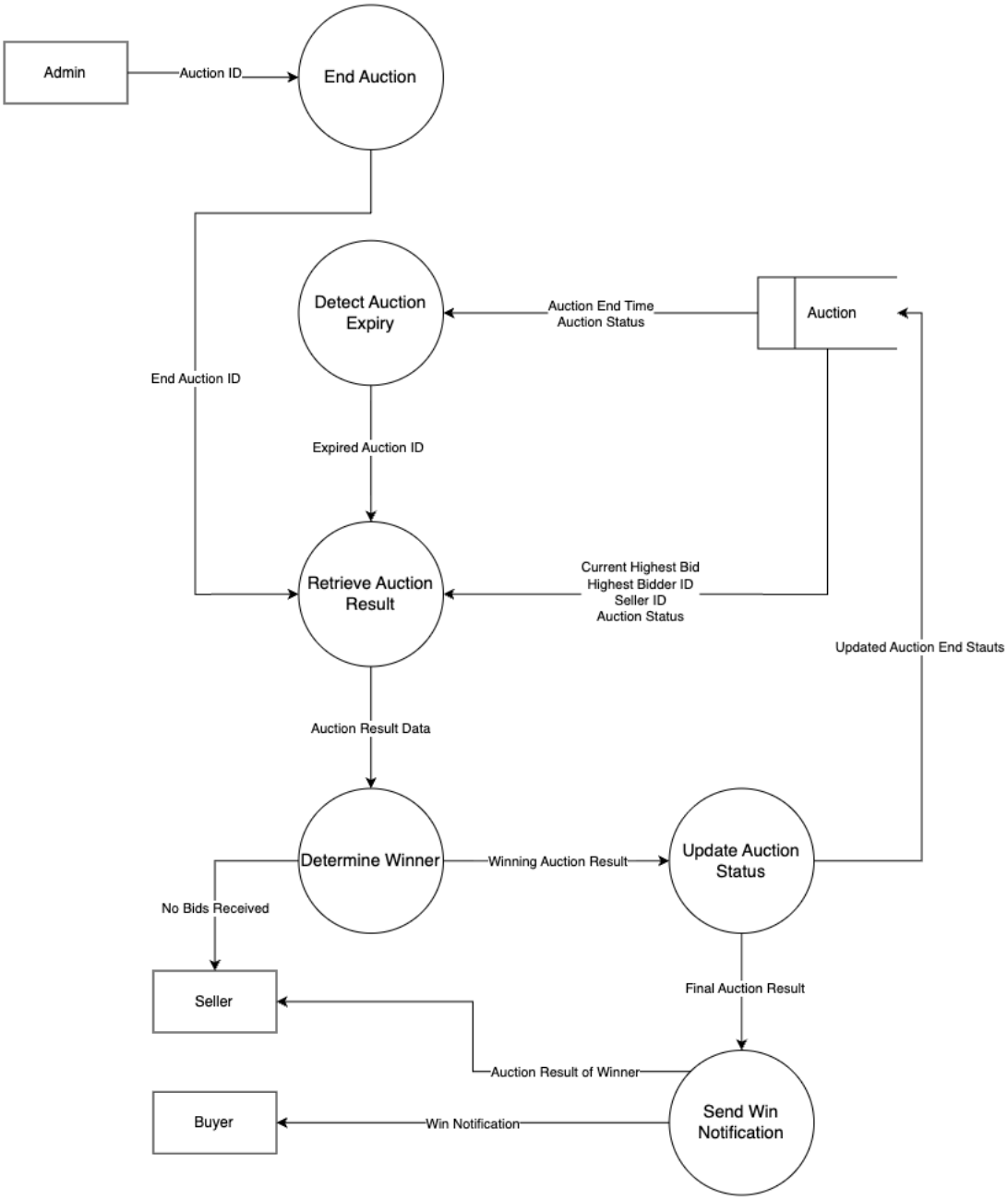

Figure 5: End Auction and Determine Winner

Requirement ID: req_buy_9

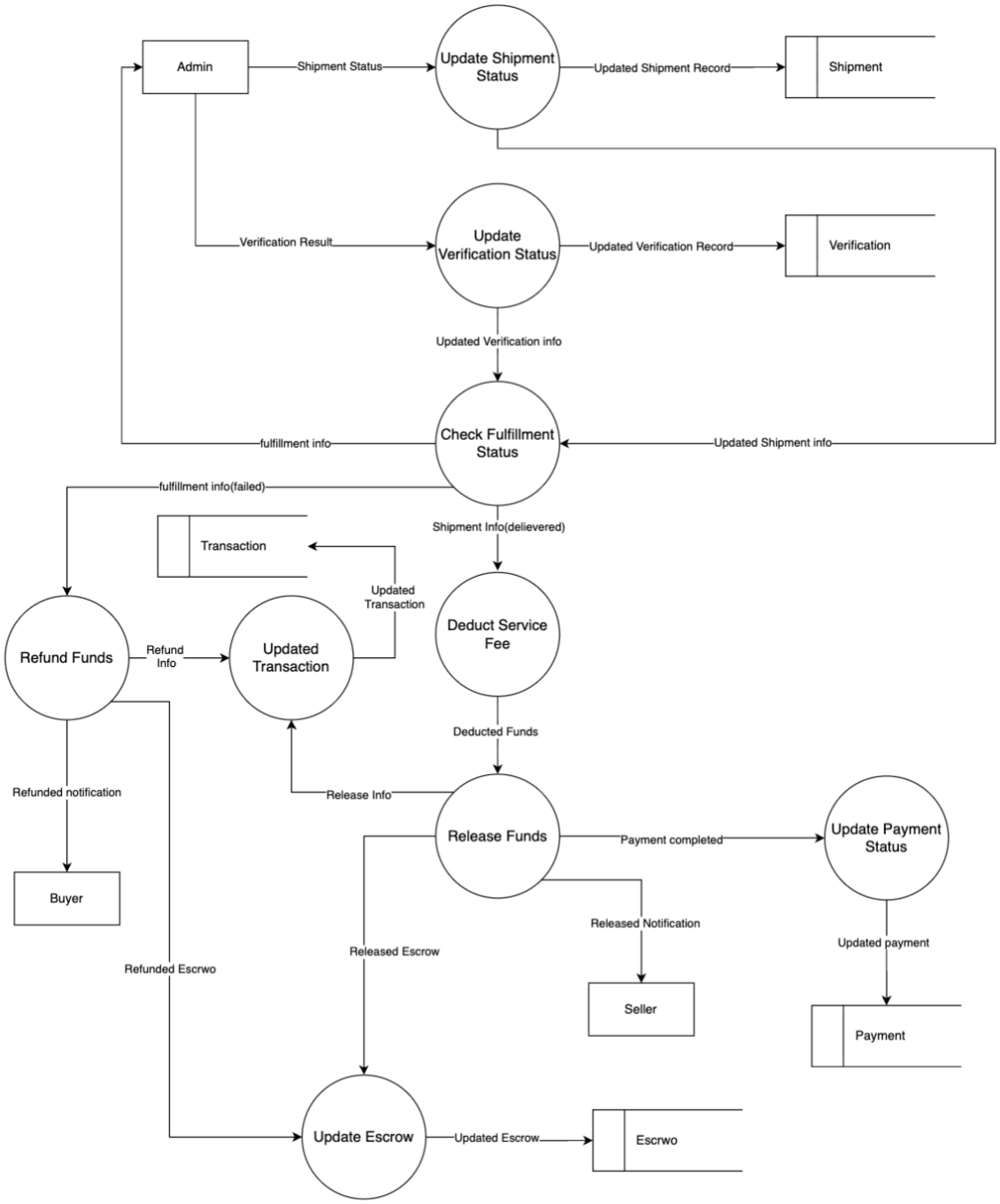

Figure 6: Update Fulfillment Status, Release and Refund

Requirement ID: req_admin_2, req_buy_3, req_buy_5

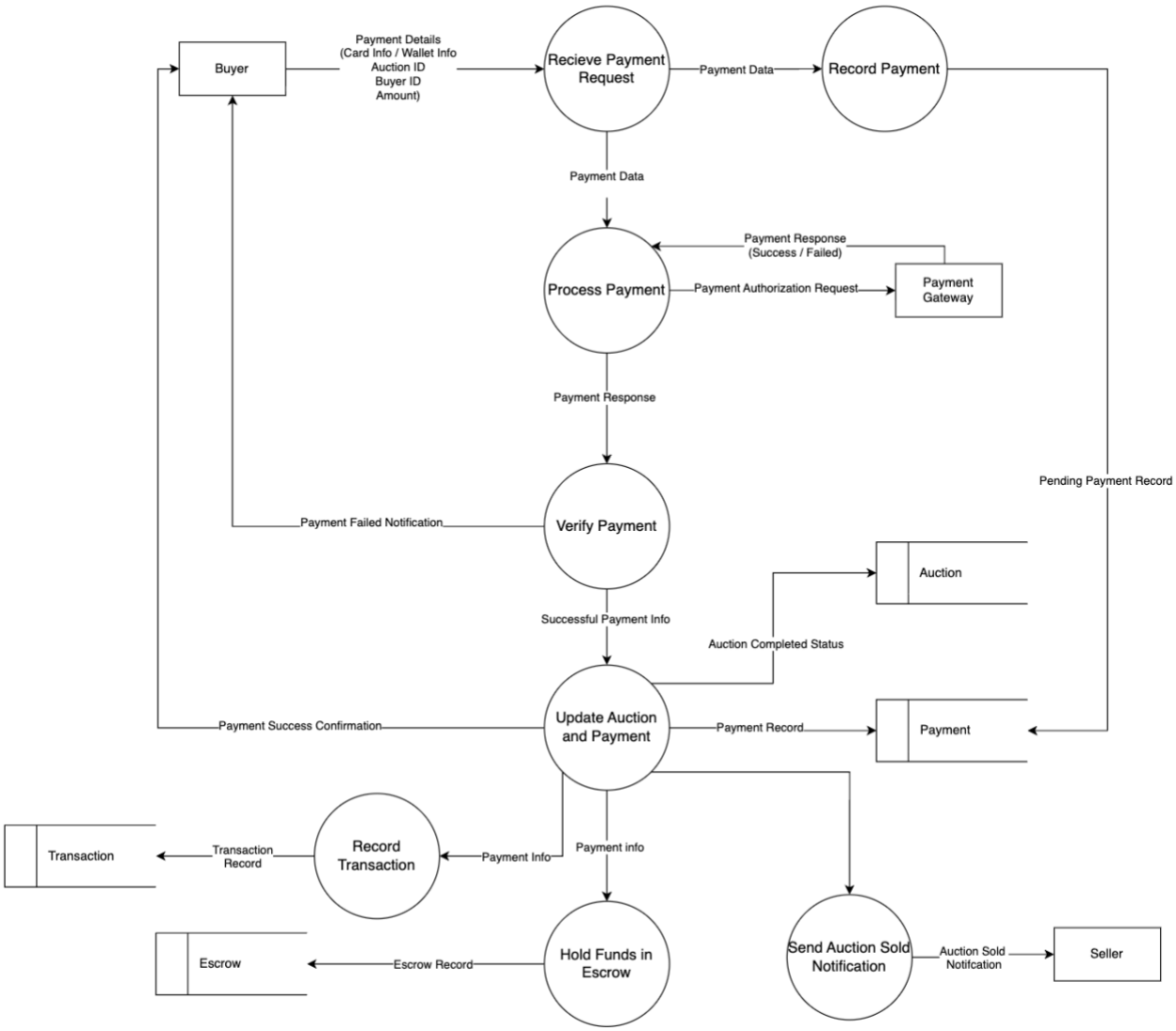

Figure 7: Manage Payment and Hold Escrow

Requirement ID: req_sell_1, req_sell_2, req_sell_4, req_admin_1, req_buy_4, req_buy_7

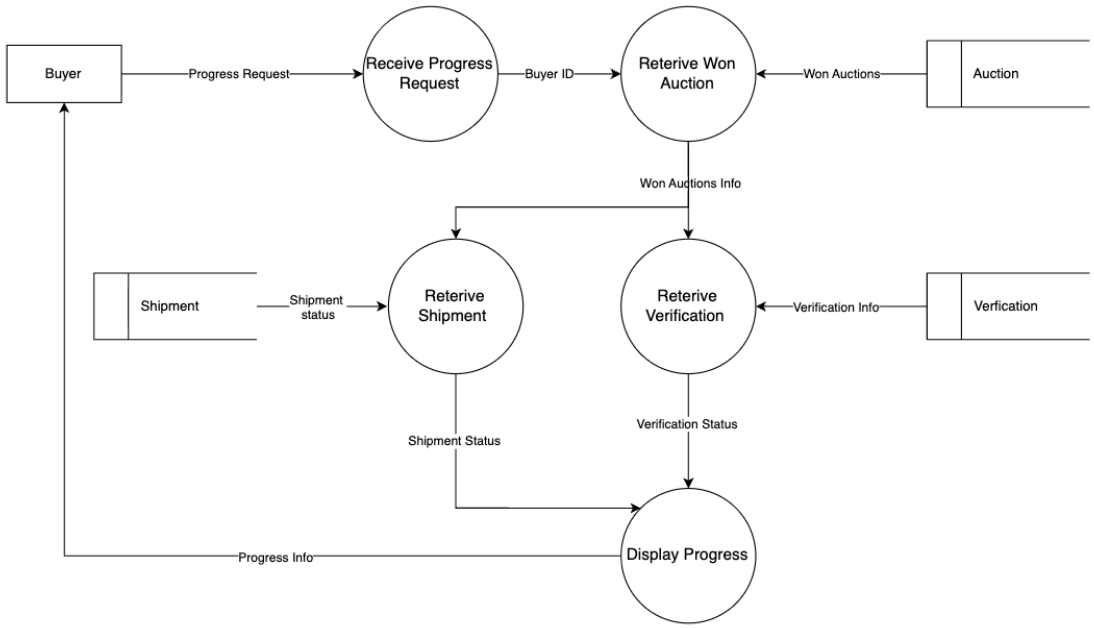

Figure 8: View Won Auctions Progress of Buyer

Requirement ID: req_buy_6

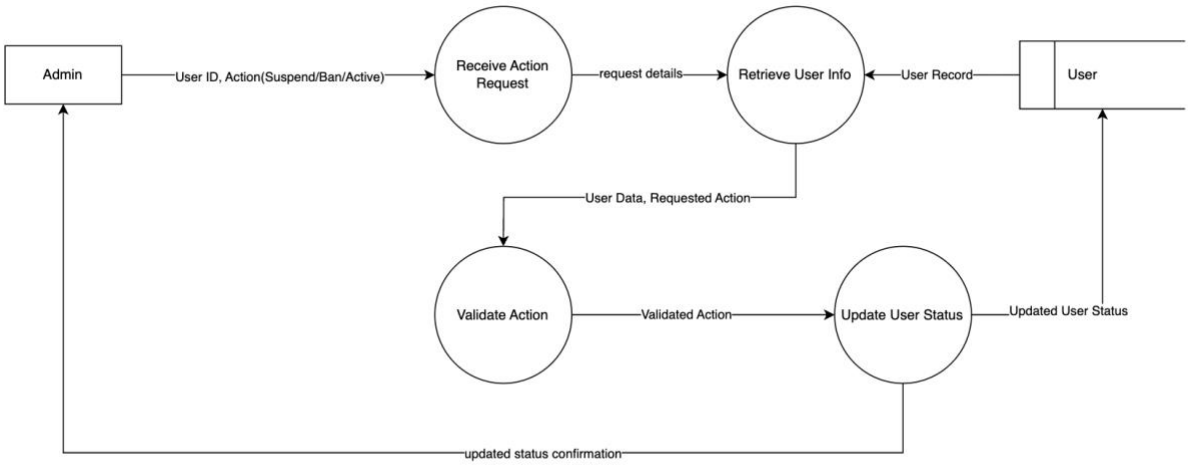

Figure 9: Suspend, Ban, Active User Account

Requirement ID: req_admin_5

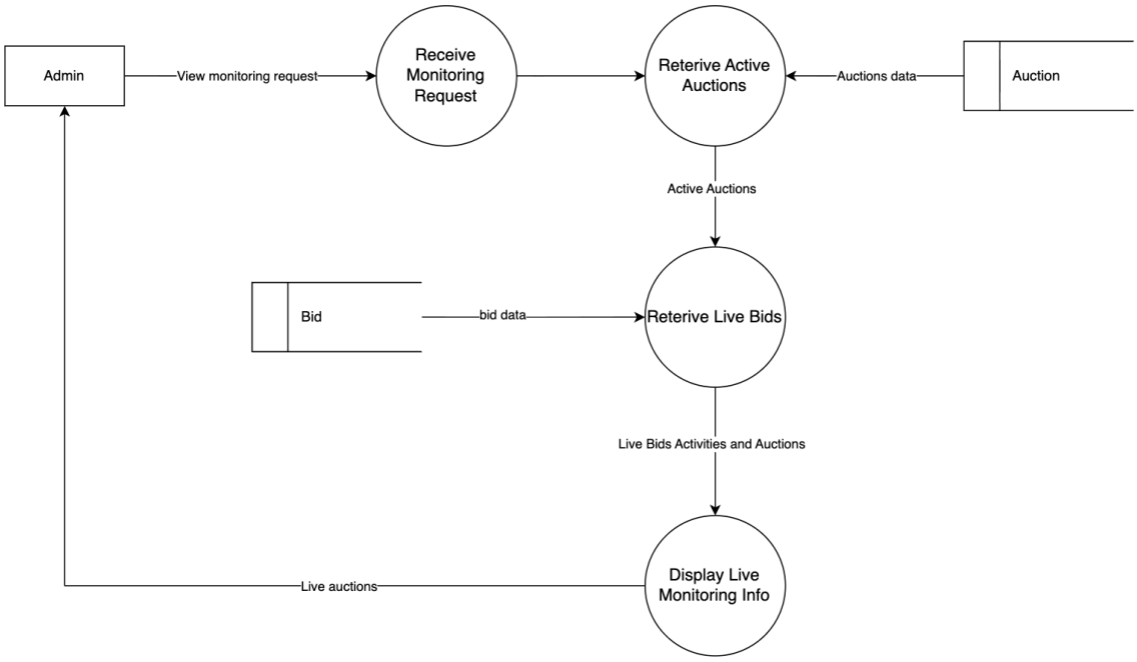

Figure 10: View Live Monitoring for Admin

Requirement ID: req_admin_3

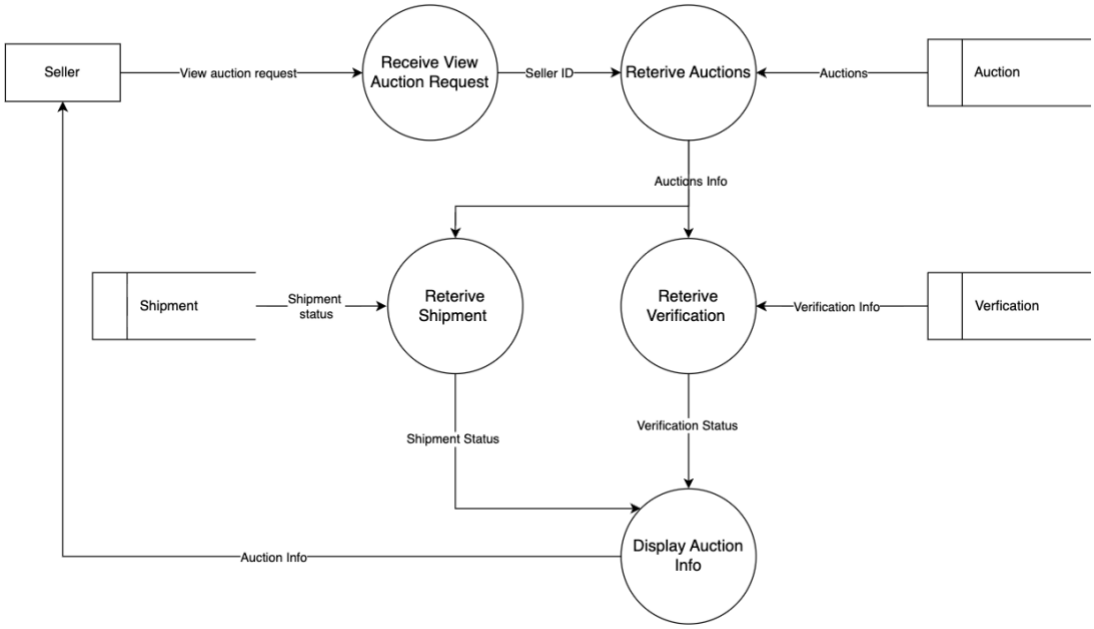

Figure 11: View Auctions for Seller

Requirement ID: req_sell_5

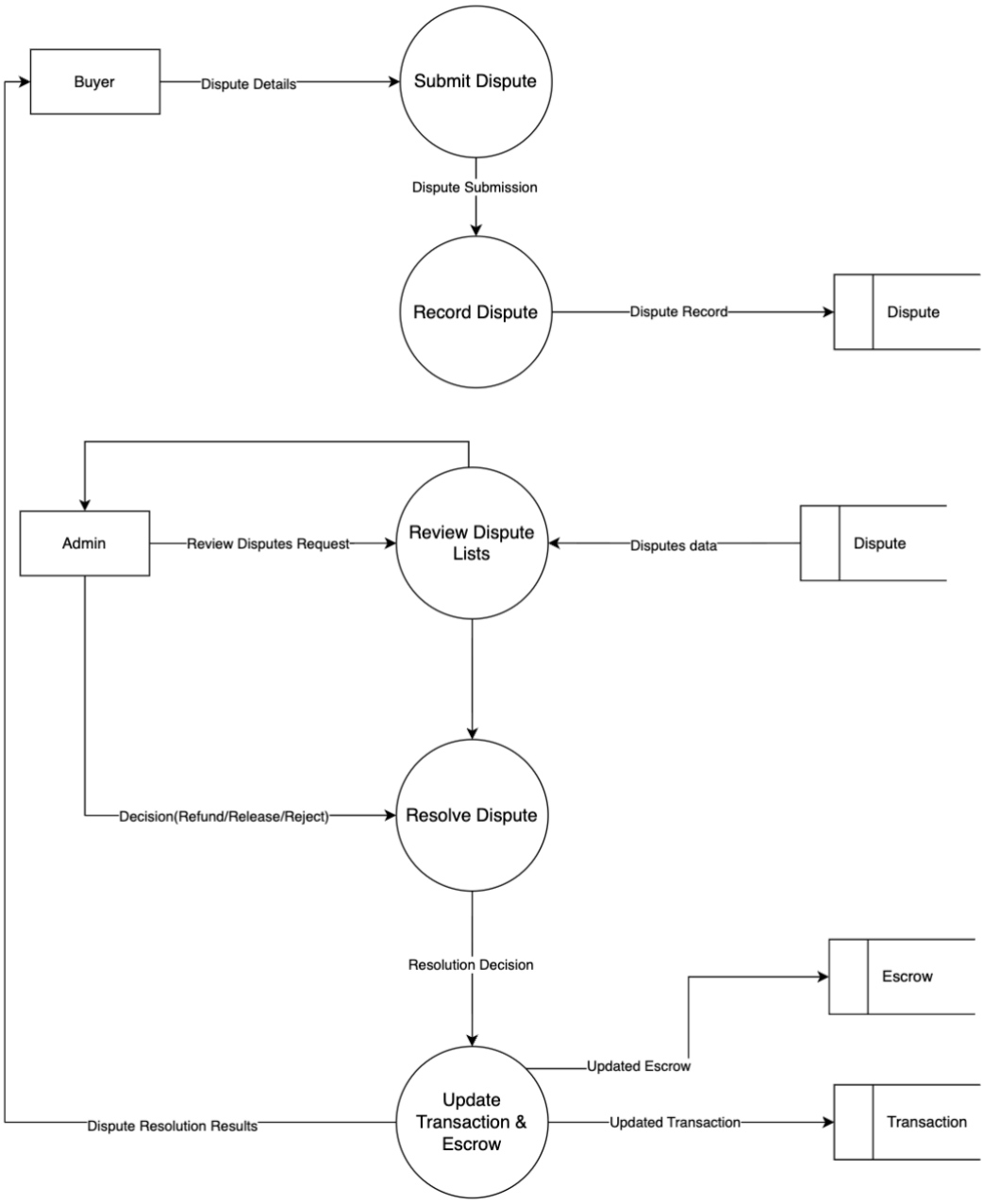

Figure 12: Submit and Resolve Dispute

Requirement ID: req_admin_4

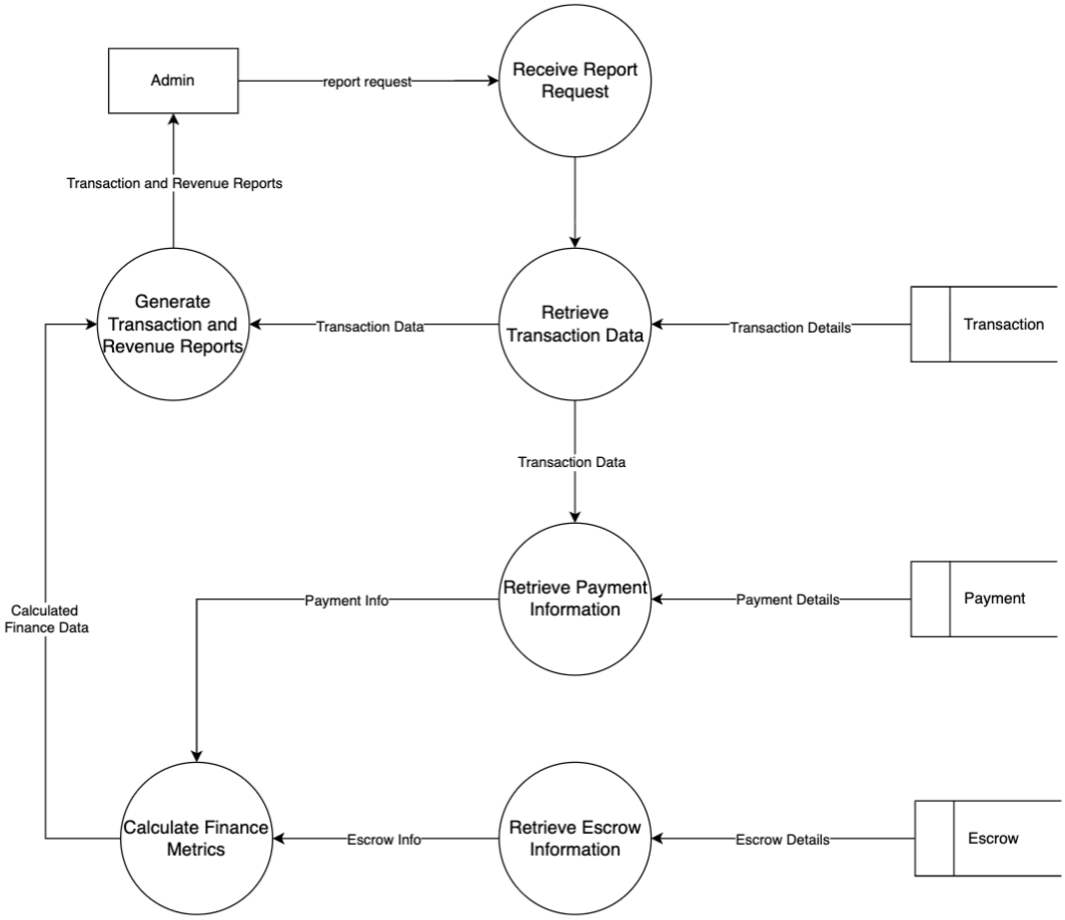

Figure 13: View and Generate Report

Requirement ID: req_admin_6

Entity Relation Diagram

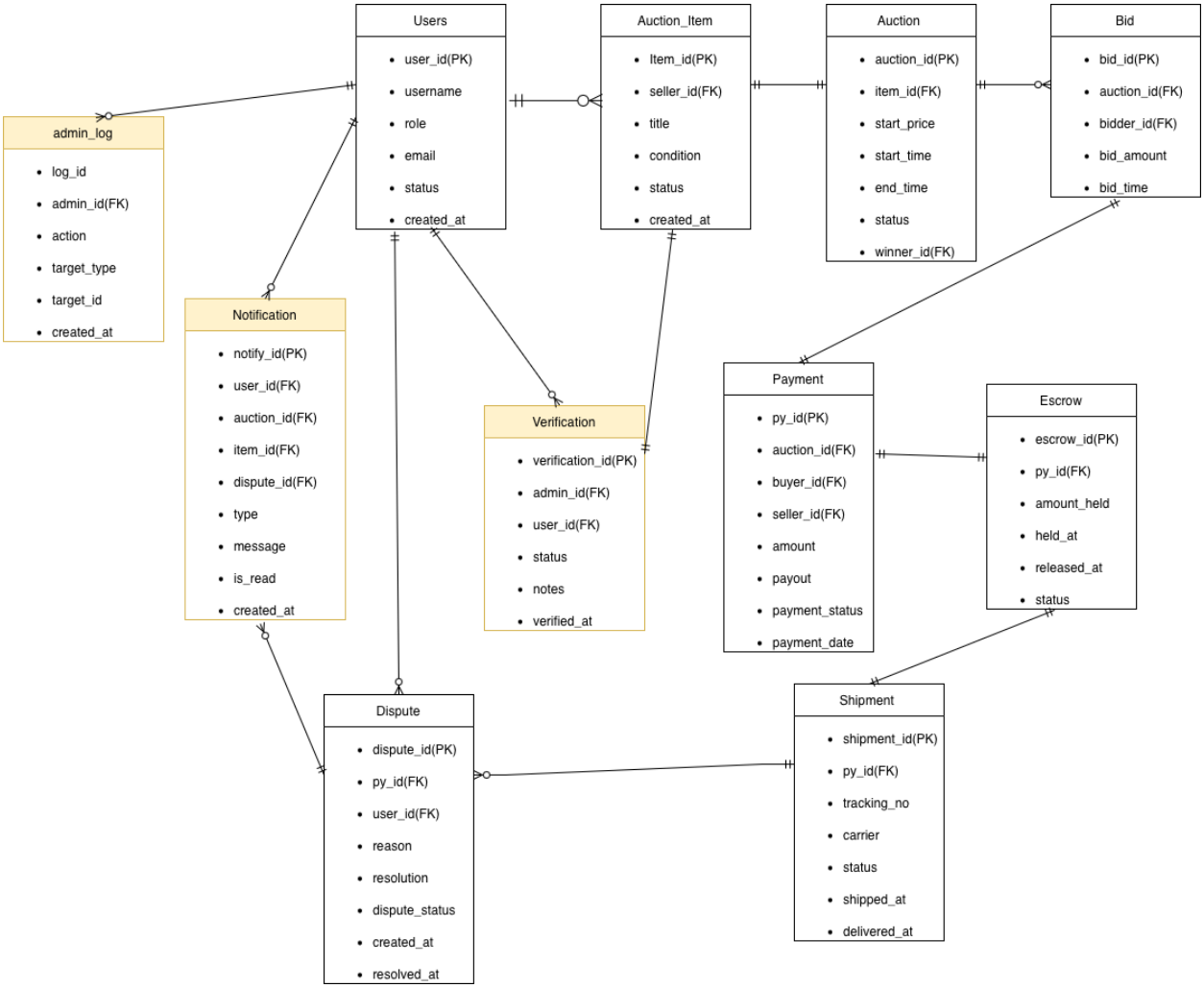

Figure 14: Entity Relation Diagram

Data Dictionary

Representative sample data illustrating how each entity is populated in the system.

users

user_id	usernamex	email	role	is_suspended (status)	created_at
1	popfan_sara	sara@email.com	buyer	FALSE	2024-11-01 09:12:00
2	collector_mia	mia@email.com	seller	FALSE	2024-11-03 14:05:00
3	rarepiece_jay	jay@email.com	seller	FALSE	2024-11-05 11:30:00
4	bidmaster_tom	tom@email.com	buyer	FALSE	2024-11-10 08:00:00

5	platform_admin	admin@popmart-auction.com	admin	FALSE	2024-10-01 00:00:00
6	frauduser_x	fraud@fake.com	buyer	TRUE	2024-11-15 16:20:00

Table 5: Data dictionary of users

auction_item

item_id	seller_id	title	condition	status	created_at
1	2	Labubu The Monsters – Tasty Macarons (Rare)	sealed	approved	2024-11-10 10:00:00
2	2	Skullpanda Exclusive Edition – Series 5	sealed	approved	2024-11-11 09:30:00
3	3	Dimoo Space Travel – Secret Figure	opened	pending_review	2024-11-12 13:15:00
4	3	Molly Zodiac Series – Dragon (Limited)	limited_edition	approved	2024-11-13 08:45:00
5	2	Crybaby × Powerpuff Girls Collaboration	sealed	sold	2024-11-07 11:00:00

Table 6: Data dictionary of auction_item

auction

auction_id	item_id	start_price (THB)	current_price (THB)	start_time	end_time	status	winner_id
1	1	1,500	4,200	2024-11-12 10:00	2024-11-15 10:00	closed	1
2	2	2,000	2,000	2024-11-13 10:00	2024-11-16 10:00	active	NULL
3	4	3,000	5,800	2024-11-14 09:00	2024-11-17 09:00	active	NULL
4	5	1,200	3,500	2024-11-08 10:00	2024-11-11 10:00	closed	4

Table 7: Data dictionary of auction

bid

bid_id	auction_id	bidder_id	bid_amount (THB)	bid_time	is_winning
1	1	1	1,800	2024-11-12 11:00:00	FALSE
2	1	4	2,500	2024-11-13 09:30:00	FALSE
3	1	1	4,200	2024-11-14 16:00:00	TRUE
4	4	4	1,500	2024-11-08 11:00:00	FALSE
5	4	1	2,800	2024-11-09 10:00:00	FALSE
6	4	4	3,500	2024-11-10 14:00:00	TRUE

Table 8: Data dictionary of bid

payment

py_id	auction_id	buyer_id	seller_id	amount (THB)	Payout (THB)	Paymet_status	Payment_date
1	1	1	2	4,200	3,780	released	2024-11-15 10:05:00
2	4	4	2	3,500	3,150	released	2024-11-11 10:05:00

Table 9: Data dictionary of payment

escrow

escrow_id	transaction_id	amount_held (THB)	held_at	released_at	status
1	1	4,200	2024-11-15 10:05:00	2024-11-20 14:00:00	released
2	2	3,500	2024-11-11 10:05:00	2024-11-17 11:30:00	released

Table 10: Data dictionary of escrow

verification

verification_id	item_id	admin_id	status	notes	verified_at
1	1	5	passed	Item is authentic and sealed. Packaging intact. Matches listing photos.	2024-11-17 10:00:00
2	5	5	passed	Authentic sealed Crybaby × PPG collaboration box. QR code verified.	2024-11-13 09:30:00

Table 11: Data dictionary of verification

shipment

shipment_id	py_id	tracking_no	carrier	status	shipped_at	delivered_at
1	1	TH7891234560	Kerry Express	delivered	2024-11-15 14:00:00	2024-11-20 12:00:00
2	2	TH3456789012	Flash Express	delivered	2024-11-11 15:00:00	2024-11-17 10:00:00

Table 12: Data dictionary of shipment

dispute

dispute_id	transaction_id	User_id	reason	resolution	status	created_at	Resolved_at
1	1	2	Buyer reported packaging was damaged on arrival.	Partial refund of 500 THB issued to buyer after admin review.	resolved	2024-11-20 15:00:00	2024-11-20 12:00:00

Table 13: Data dictionary of dispute

notification

notif_id	user_id	auction_id	Item_id	dispute_id	type	message	is_read	created_at
1	1	1	3	1	outbid	You have been outbid on Labubu Macarons. Current price: 2,500 THB.	TRUE	2024-11-13 09:31:00
2	1	2	2	1	auction_won	Congratulations! You won Labubu Macarons at 4,200 THB.	TRUE	2024-11-15 10:06:00
3	2	3	1	2	item_sold	Your item Labubu Macarons has been sold for 4,200 THB.	TRUE	2024-11-15 10:06:00
4	1	4	5	3	shipment_update	Your item is in transit. Tracking: TH7891234560 via Kerry Express.	TRUE	2024-11-16 08:00:00
5	2	5	4	3	payment_released	Payment of 3,780 THB has been transferred to your account.	FALSE	2024-11-20 14:01:00

Table 14: Data dictionary of notification

admin_log

log_id	admin_id	action	target_type	target_id	created_at
1	5	approved_listing	auction_item	1	2024-11-11 09:00:00
2	5	approved_listing	auction_item	2	2024-11-12 09:15:00
3	5	verified_item	auction_item	1	2024-11-17 10:00:00
4	5	released_escrow	transaction	1	2024-11-20 14:00:00
5	5	suspended_user	user	6	2024-11-15 17:00:00
6	5	resolved_dispute	dispute	1	2024-11-21 10:30:00

Table 15: Data dictionary of admin_log

USER INTERFACE

UI/UX Design

The Pop Mart Auction Platform utilizes a design focused on transparency and security. The UI ensures simple navigation to reduce complexity, while the UX provides real-time updates to build user trust during the bidding and verification process.

Prototype User Workflow

1. Landing Page

The Landing Page serves as the entry point for all users, showcasing featured Pop Mart collectibles and active auctions. It provides a clear overview of how the platform's verification and escrow system works to build immediate trust with new visitors.

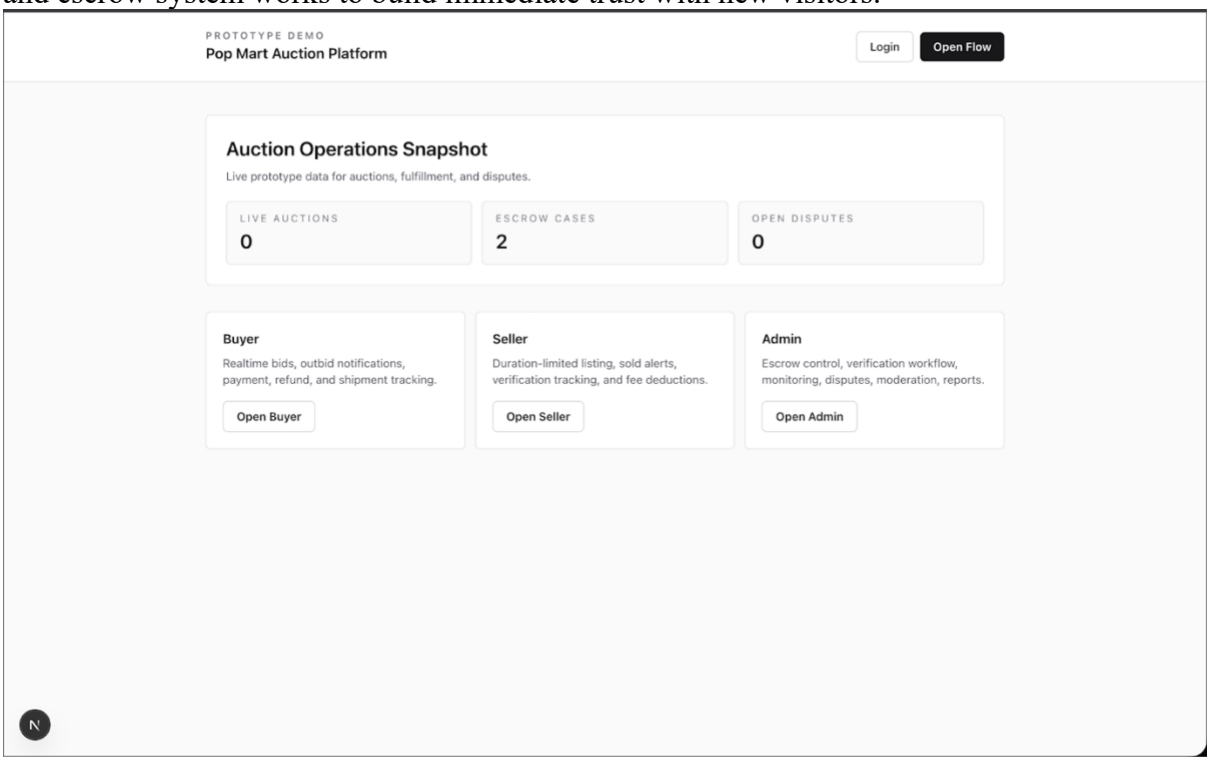

Figure 15: Landing Page

2. Sign-In Page

It includes demo controls for testing purposes and allows new users to register as either a buyer or a seller. Secure login is required to access personalized features like bidding history and profile settings.

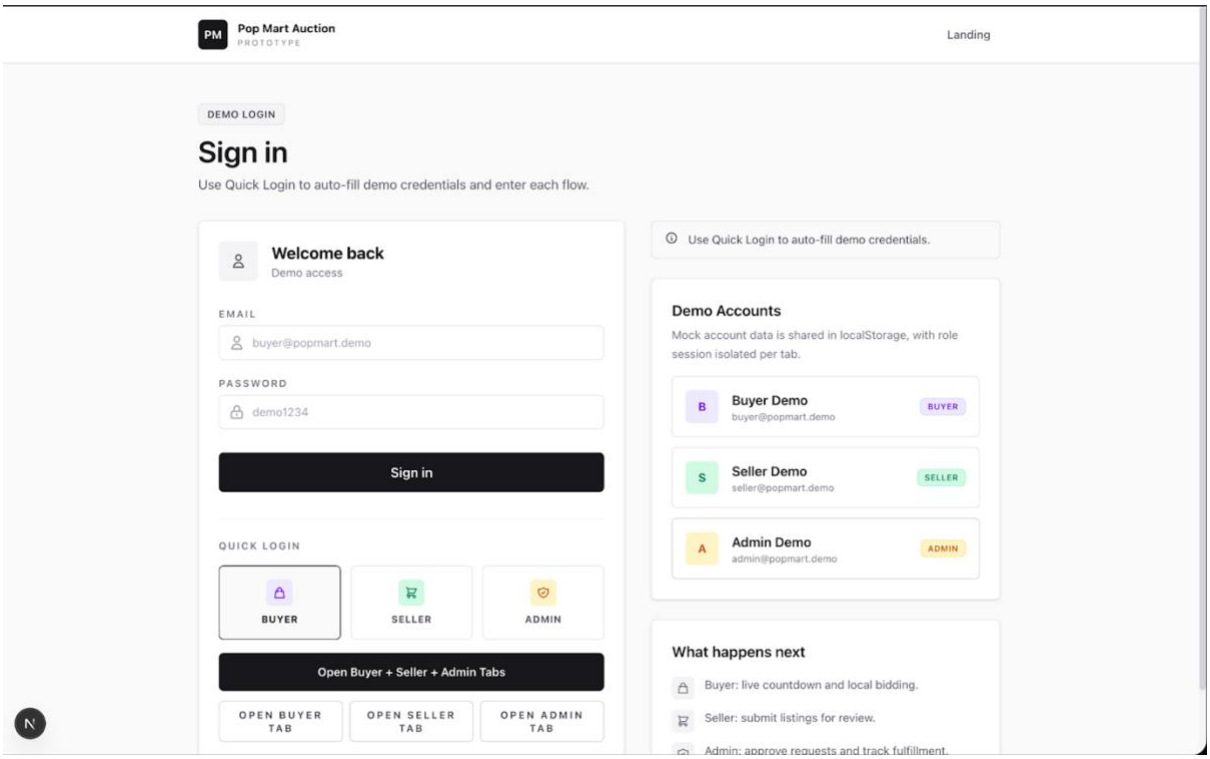

Figure 16: Sign-in page with demo controls

3. Seller Dashboard

Sellers use this page to list their Pop Mart items for auction. They can upload photos, write descriptions, and set auction durations within platform limits.

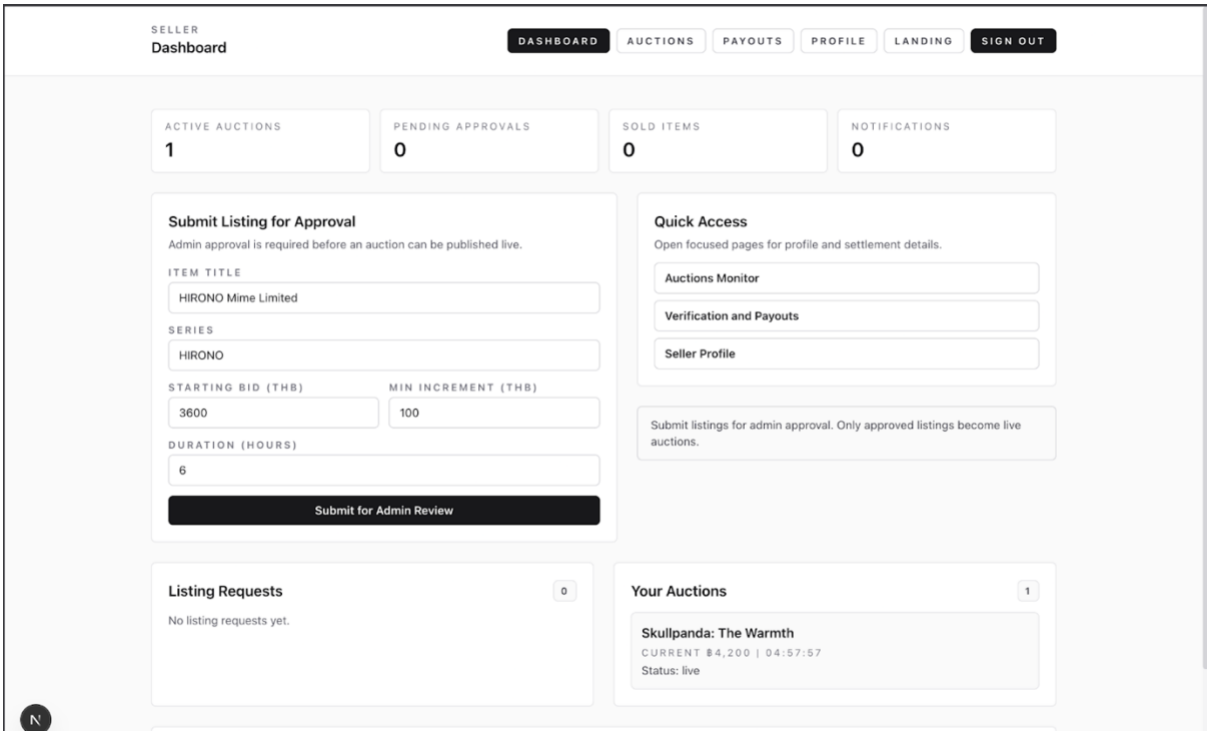

Figure 17: Seller Dashboard for req_sell_3

Once submitted, the item is marked as "Pending Review" until an admin approves it.

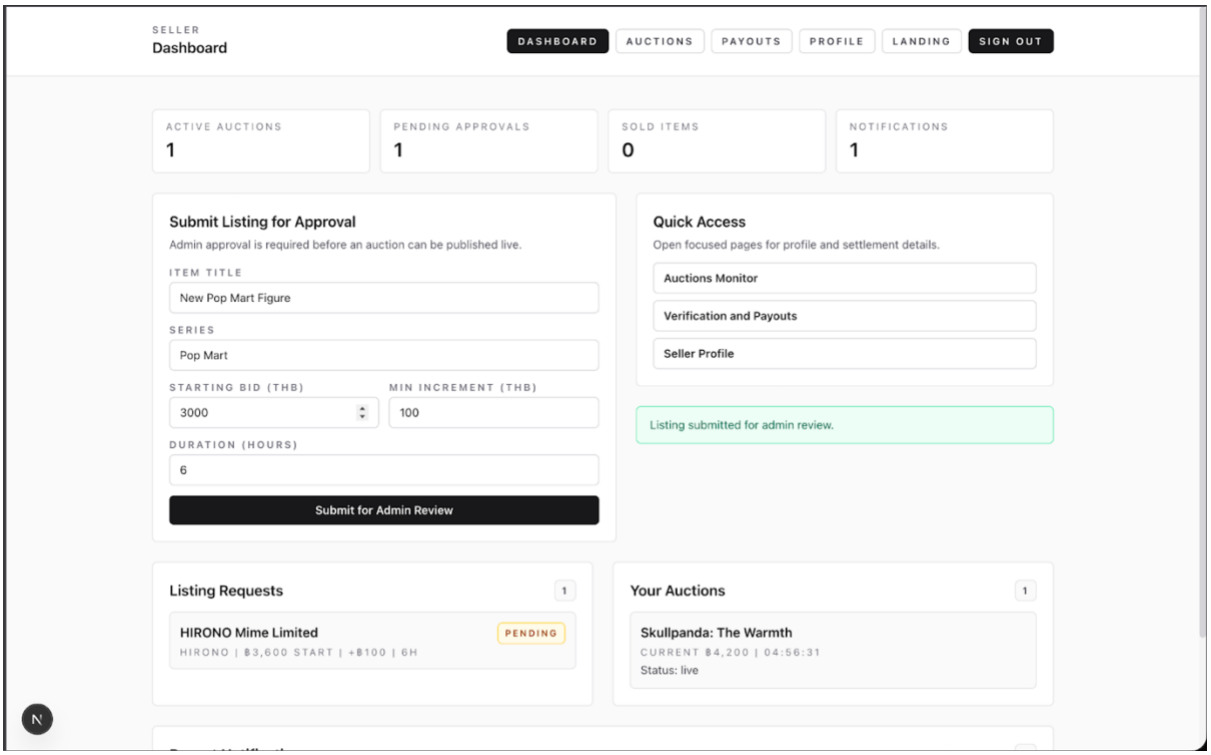

Figure 18: Seller submitted auction item for admin review

4. Admin Dashboard

The Admin Dashboard provides a high-level view of platform activity. Monitoring section allows administrators to actively track live auctions and bidding patterns in real-time to prevent suspicious activity or fraud.

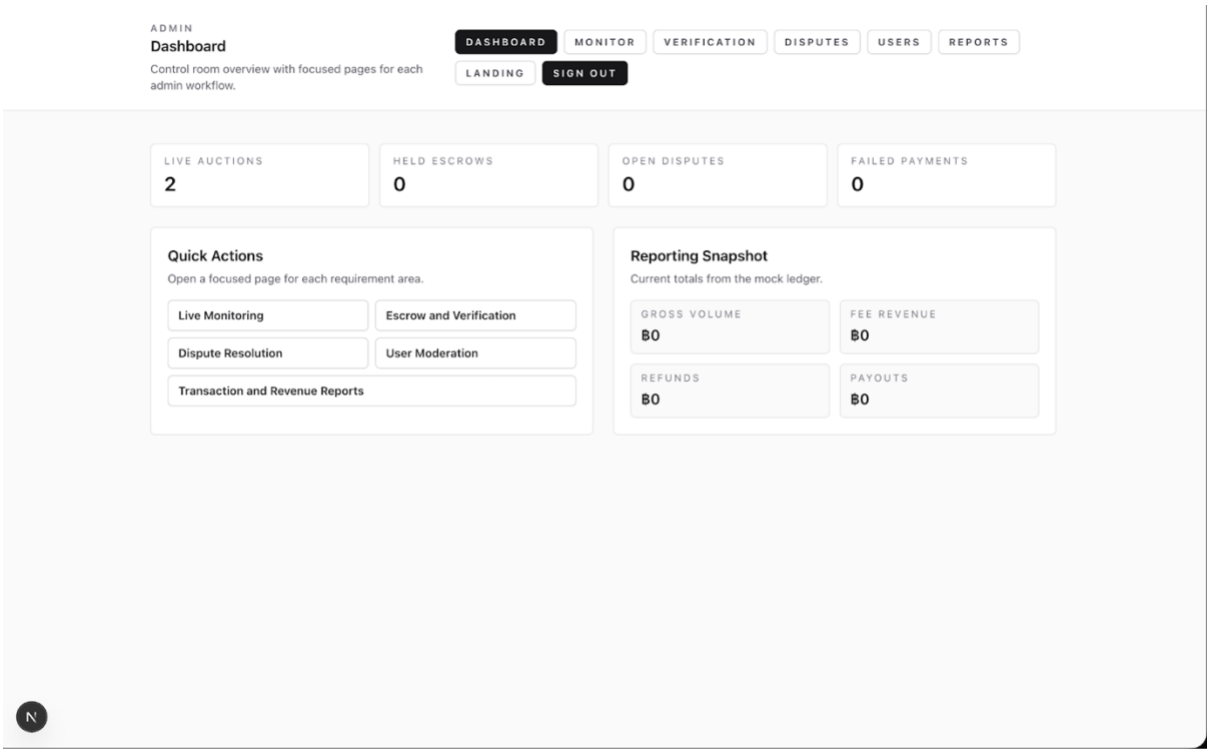

Figure 19: Admin Dashboard

5. Admin Listing Approval & Rejection Page

Administrators screen seller submissions. They can approve an item to make it go live or reject.

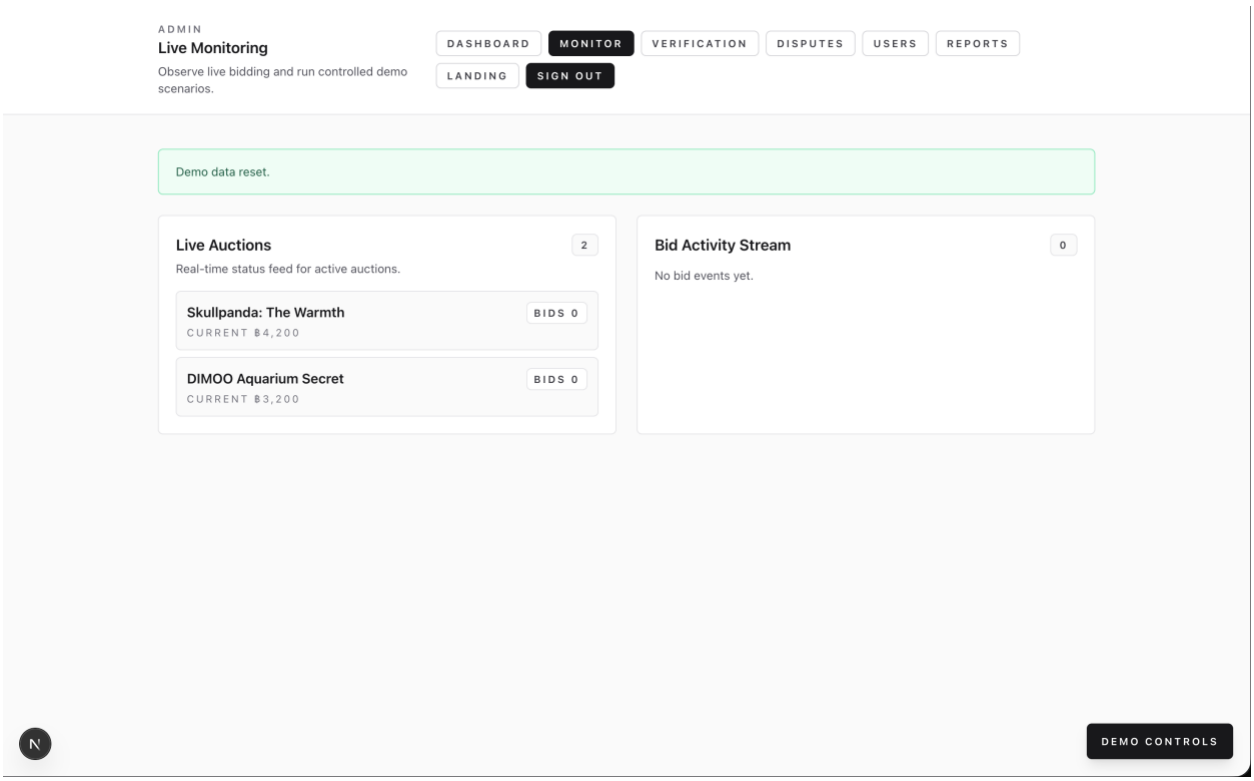

Figure 20: Admin Dashboard checking live auctions for req_admin_3

Admin page for verification of auction items and payment processing cases for sold items in auction. Admin can either approve or reject the auction item for listing.

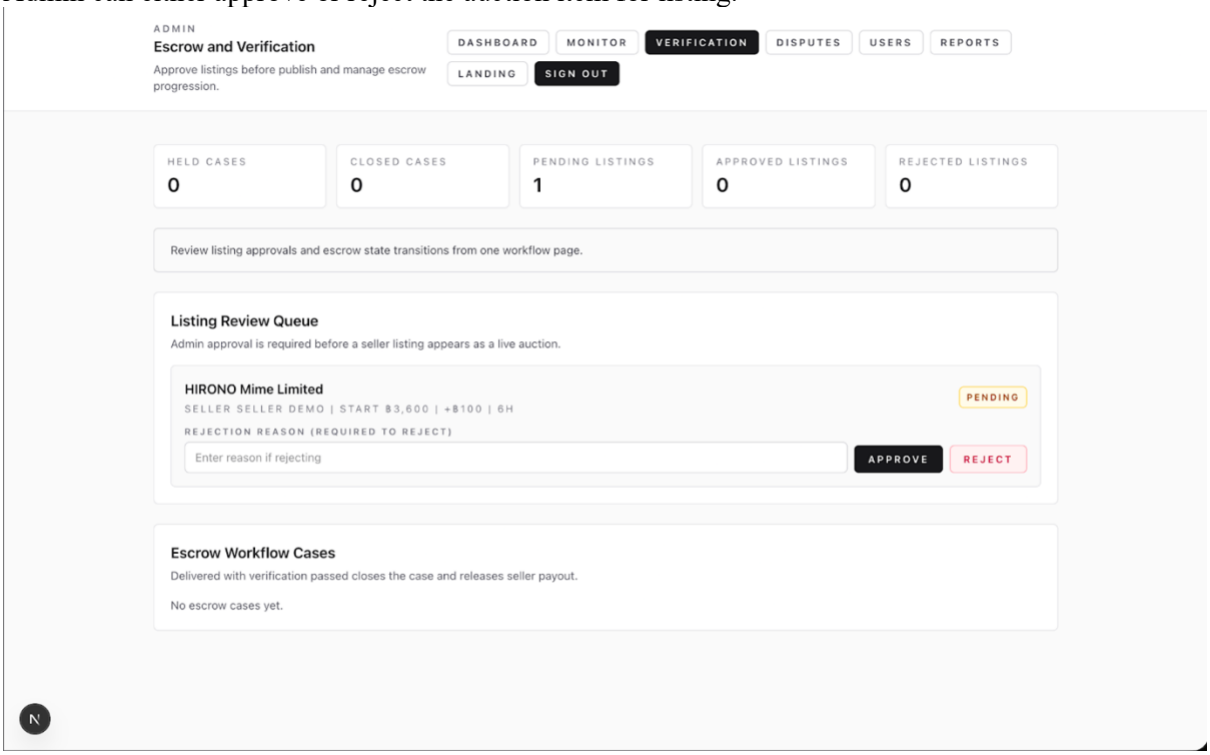

Figure 21: Verification and payment process for req_admin_7 and req_admin_8

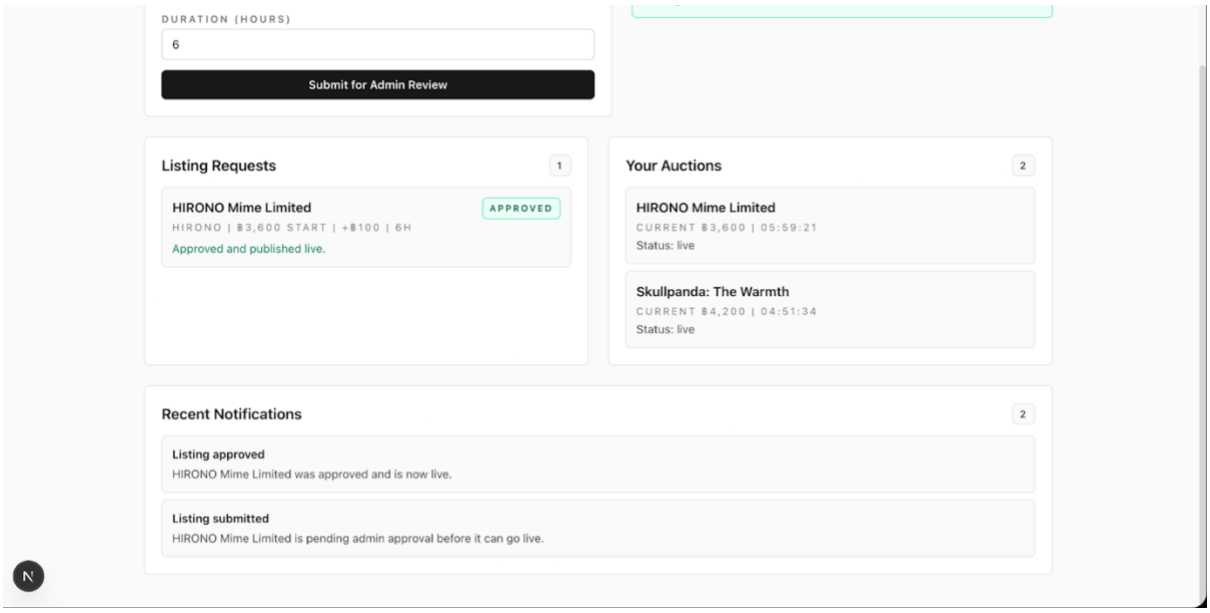

Figure 22: Seller's auction item goes live if approved

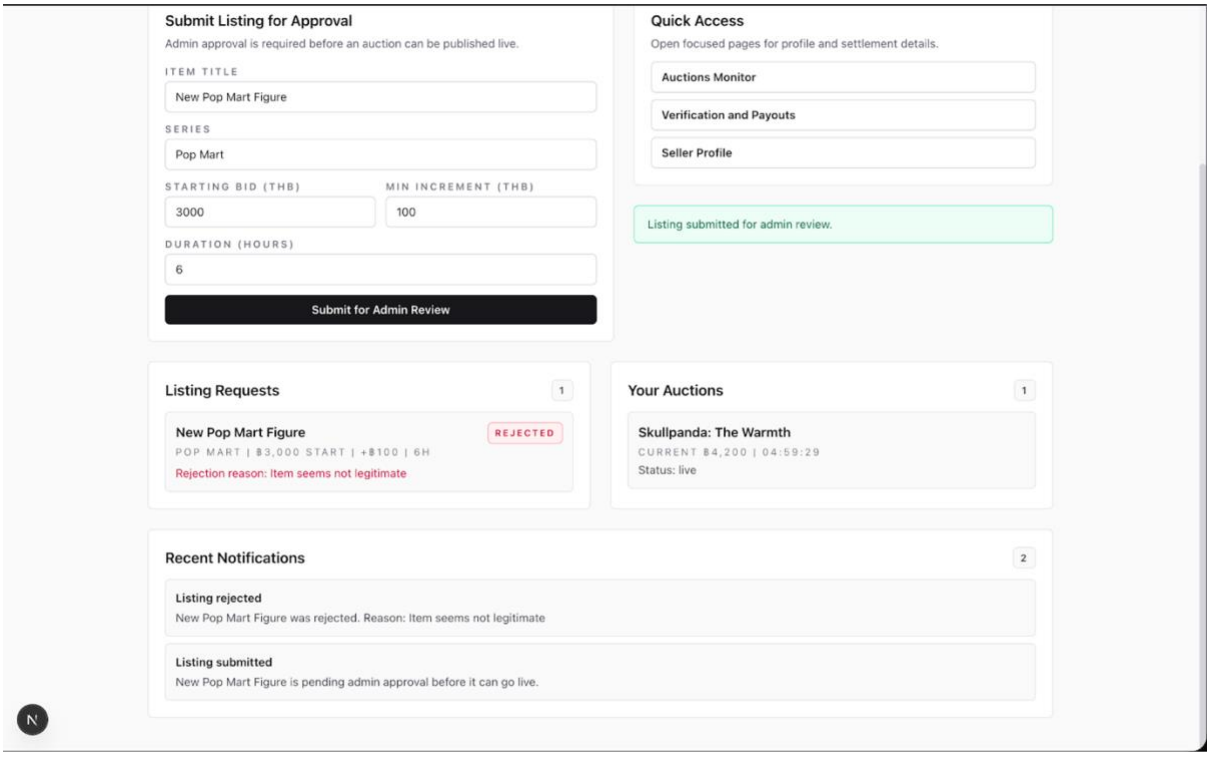

Figure 23: Seller get reason if rejected auction item for req_admin_8

6. Live Auction Page (Buyer Interface)

This is the primary page for buyers to participate in the market. It features real-time bidding functionality, allowing buyers to place bids instantly.

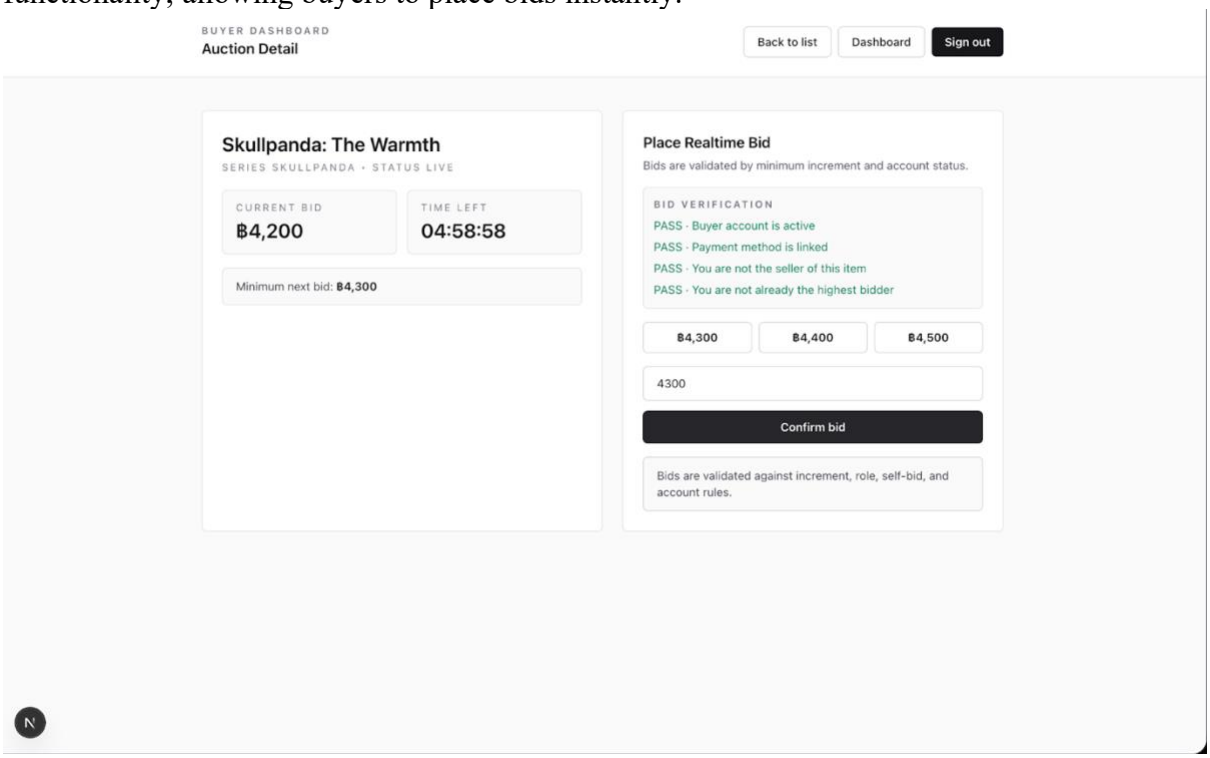

Figure 24: Bid auction item from buyer for req_buyer_1

7. Notification System

The system sends automated alerts to keep users informed. Buyers receive outbid notifications immediately so they can choose to bid again. Sellers are notified as soon as their item is sold, and both parties receive updates regarding shipment and payment releases.

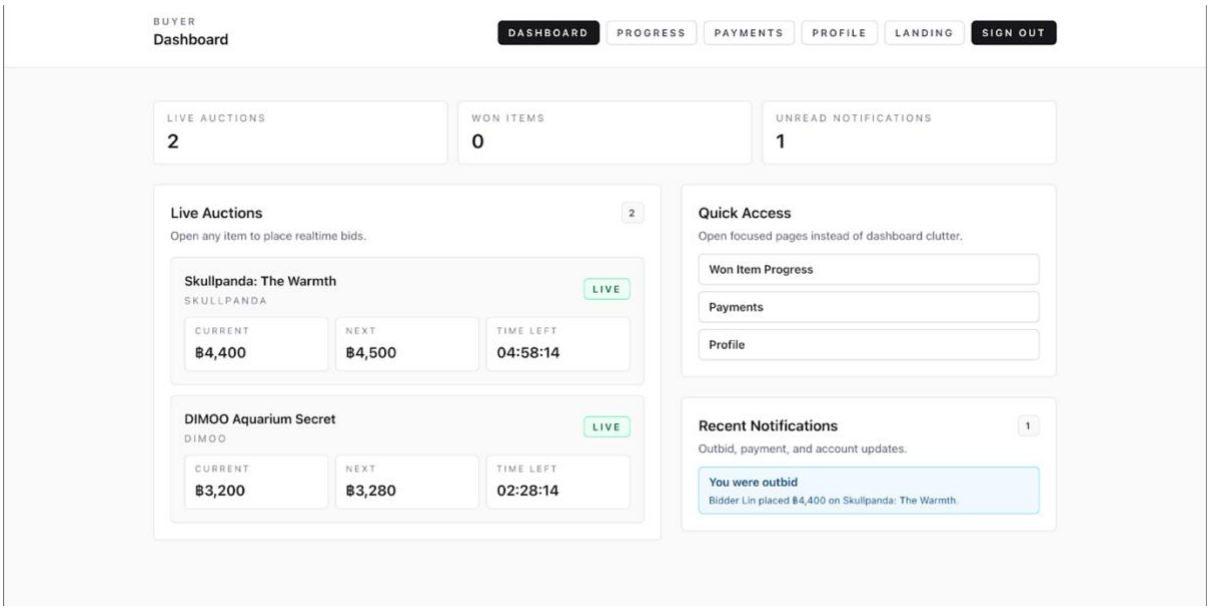

Figure 25: Show notification if buyers got outbid for req_buy_2

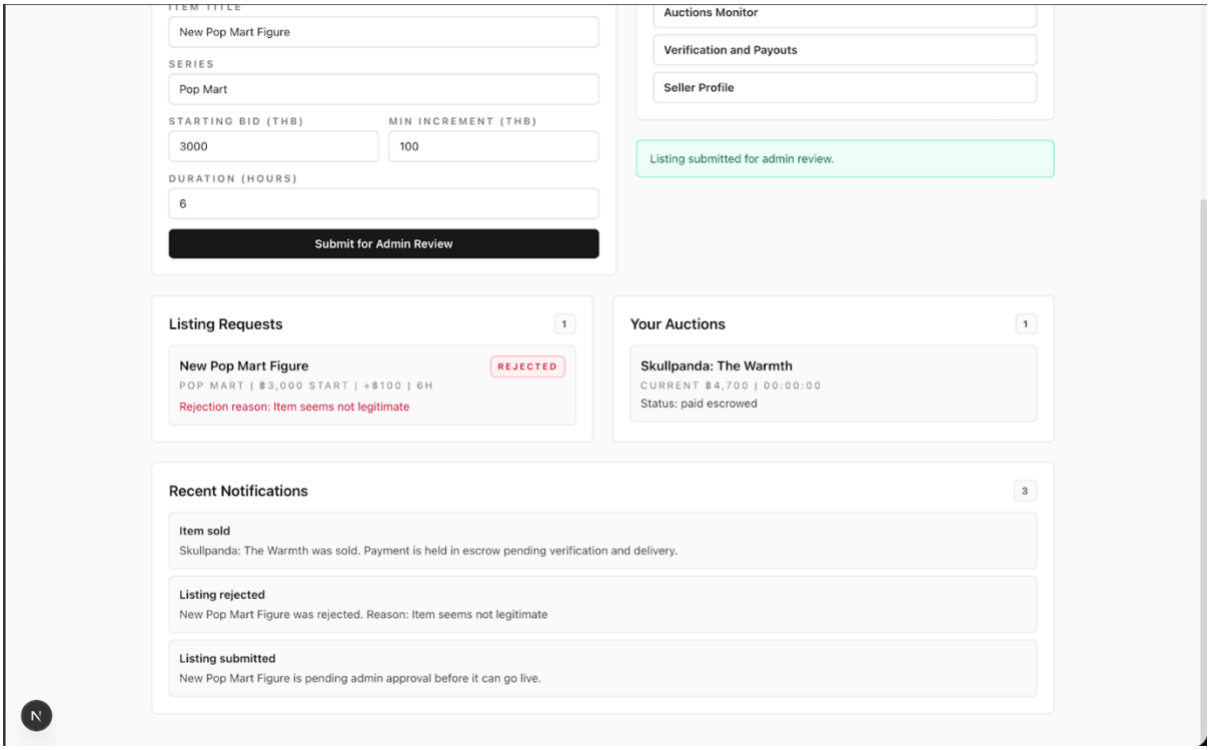

Figure 26: Sellers get notification if the item is sold for req_sell_4

8. Tracking Page for live item of buyers interface

Once an auction is won, the buyer is directed here to see the real-time progress of their item.

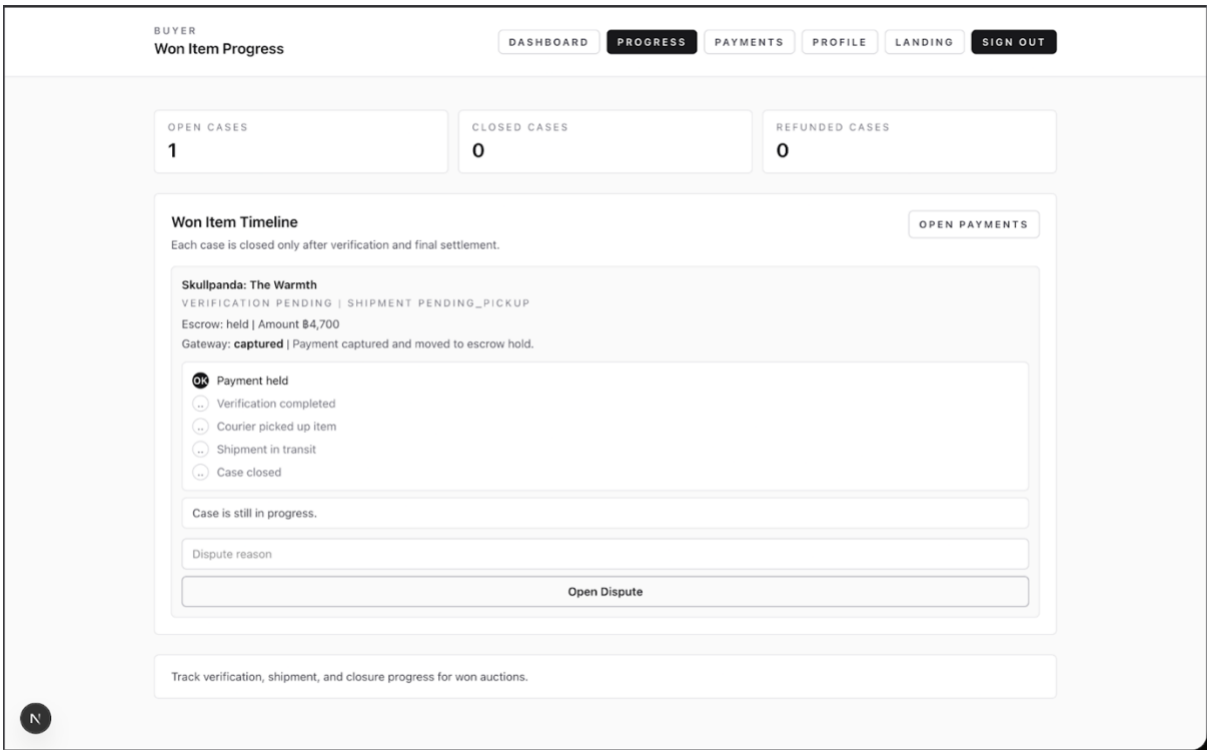

Figure 27: process for buyers to verify item from admin for req_buy_4 and req_buy_3

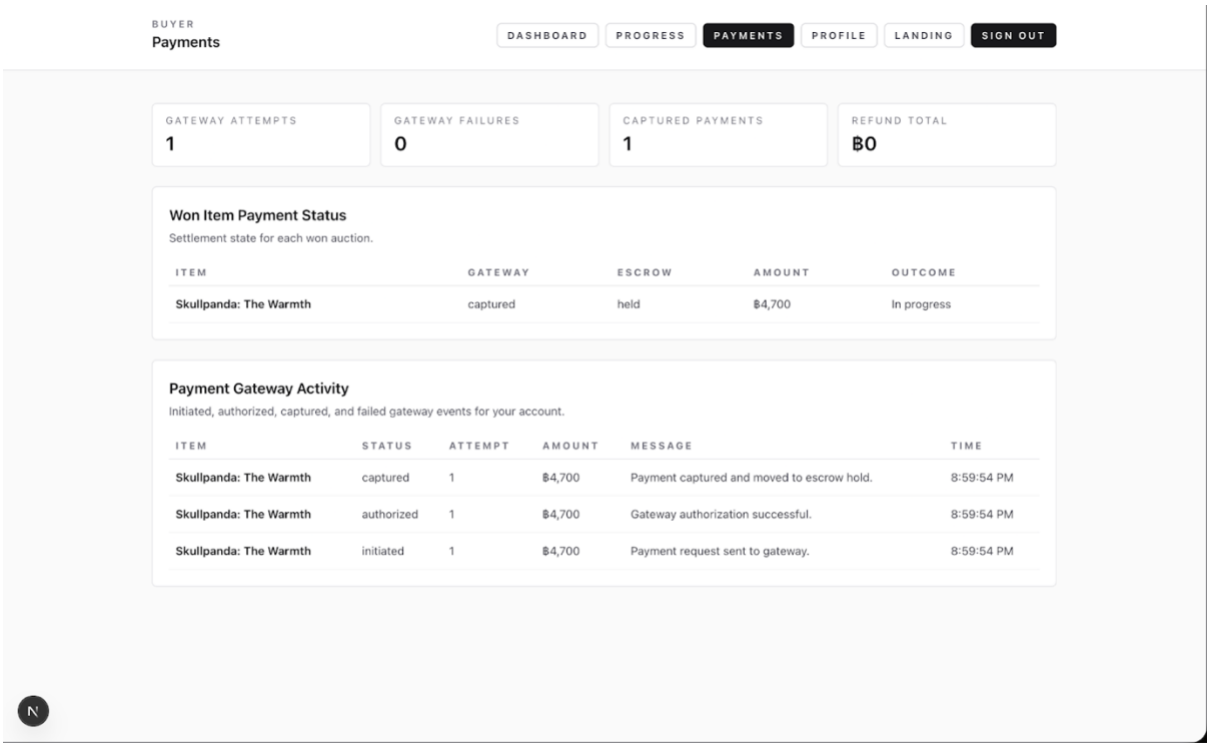

Figure 28: Tracking process for payment of auction item for req_buy_4 and req_buy_8

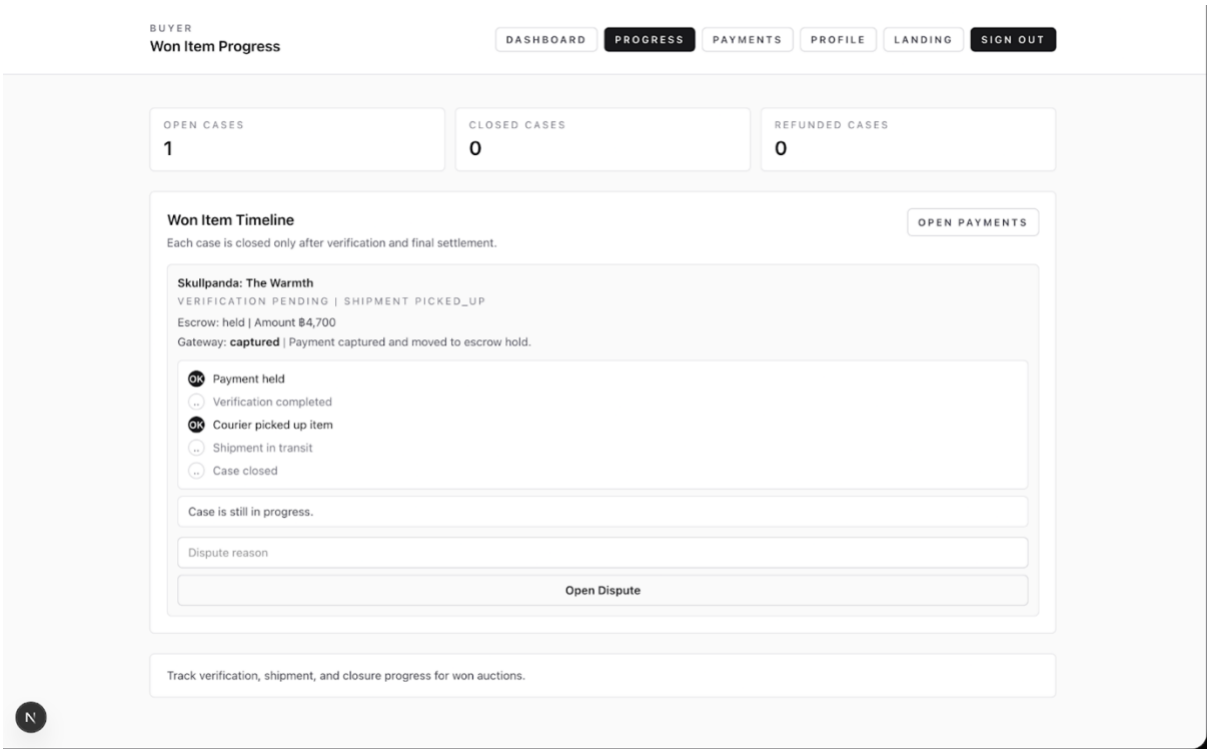

Figure 29: Progress updated for winning items buyer side for req_buy_6

9. Admin Verification and Payment Workflow

Administrators use it to log the results of the physical authenticity check. If the item passes, they trigger the shipment; if it fails, the system automatically initiates a refund to the buyer.

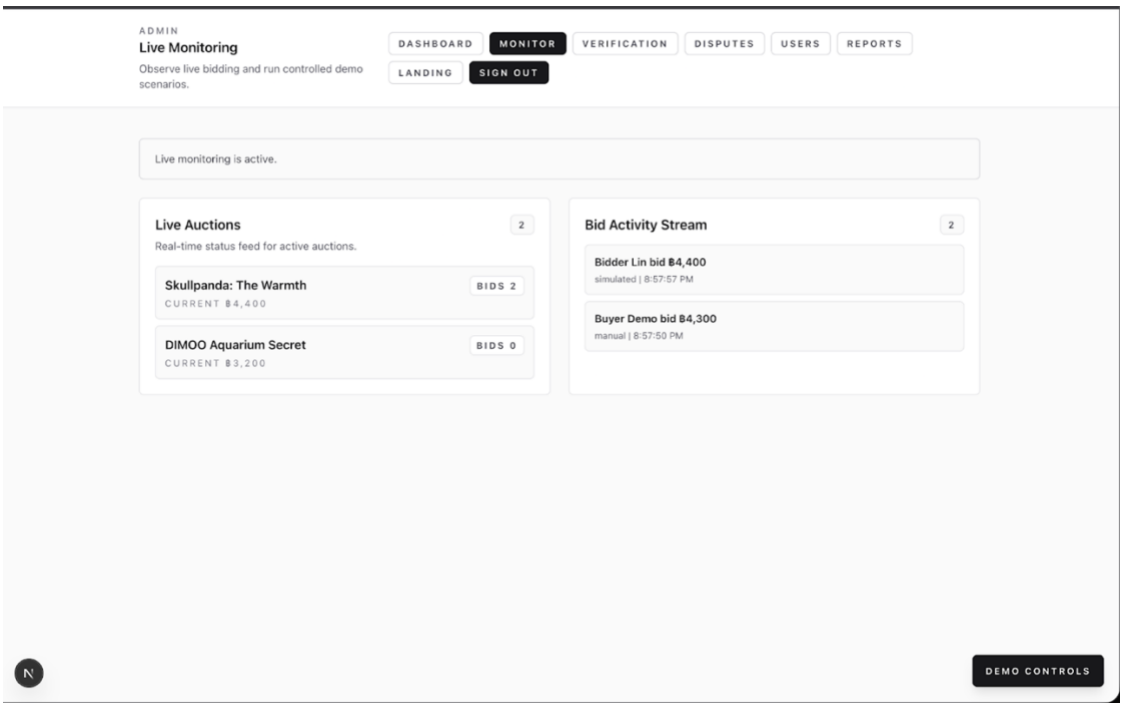

Figure 30: Admin view to monitor live auction activity for req_admin_3

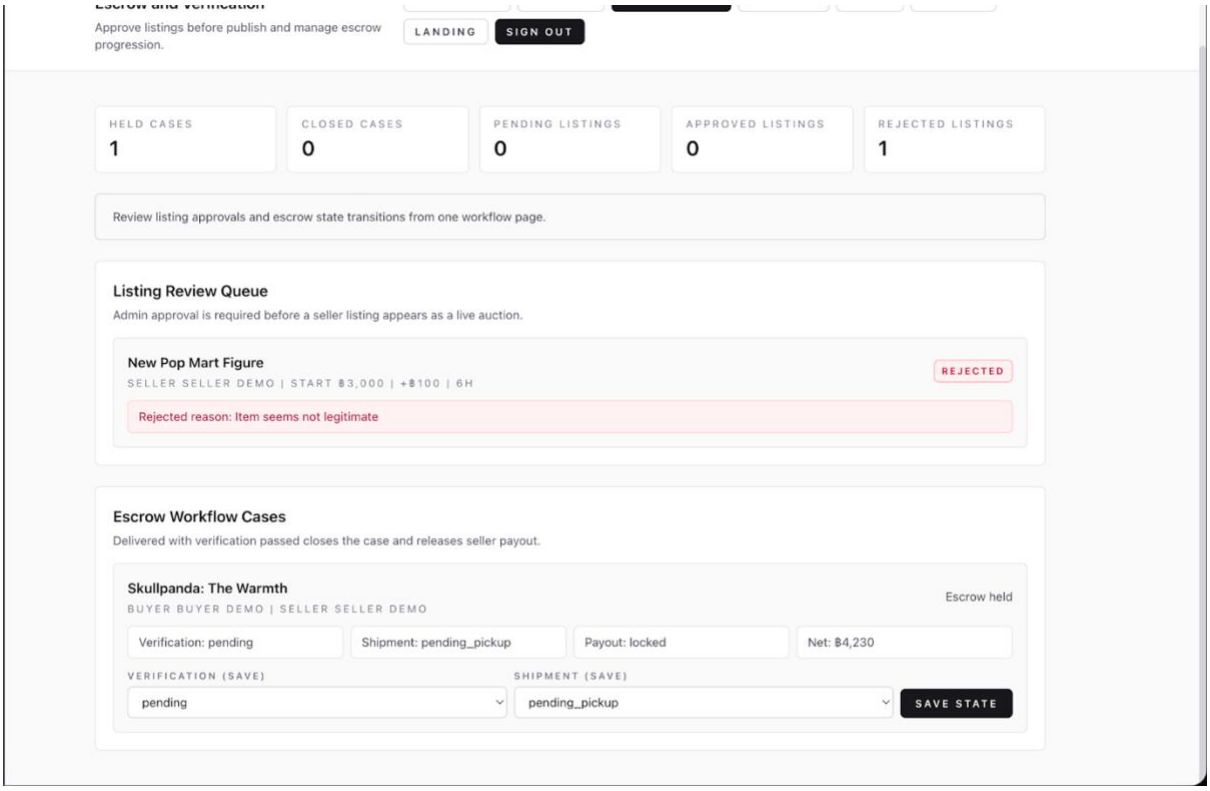

Figure 31: Admin view to monitor bidding succeed process for req_admin_1

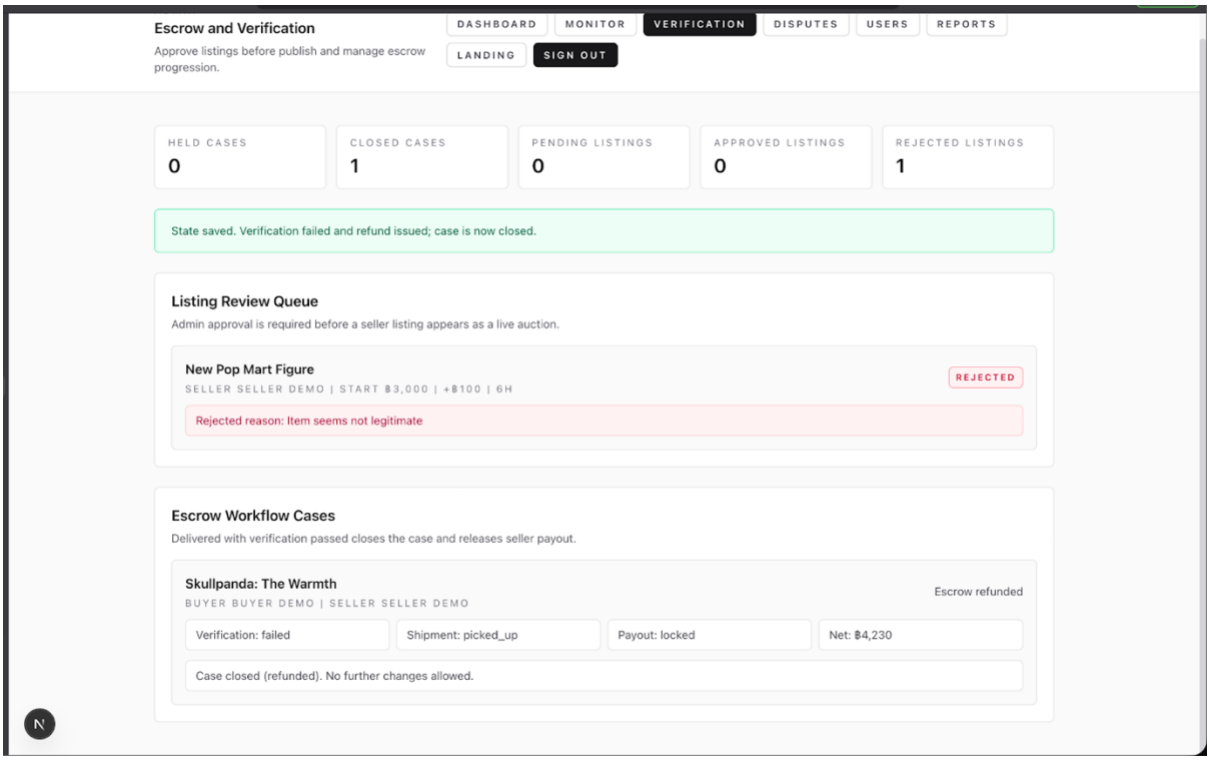

Figure 32: Admin Workflow to rejection authenticity of item for req_admin_2

10. Payment Status and Error Handling

If an item fails, the admin verification it will be refunded to the buyer. When the buyer wins the auction, but they don't have enough funds in their account then the payment will be marked as fail until they add enough money.

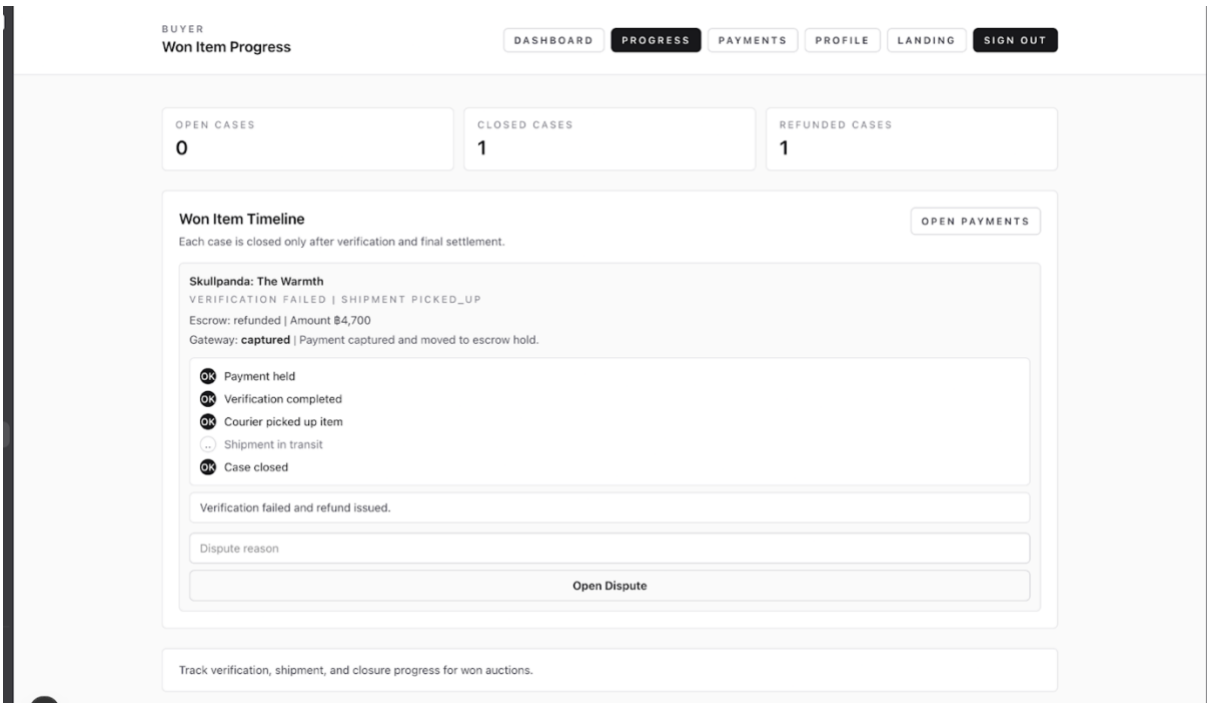

Figure 33: Refunding process for buyers for req_buy_5

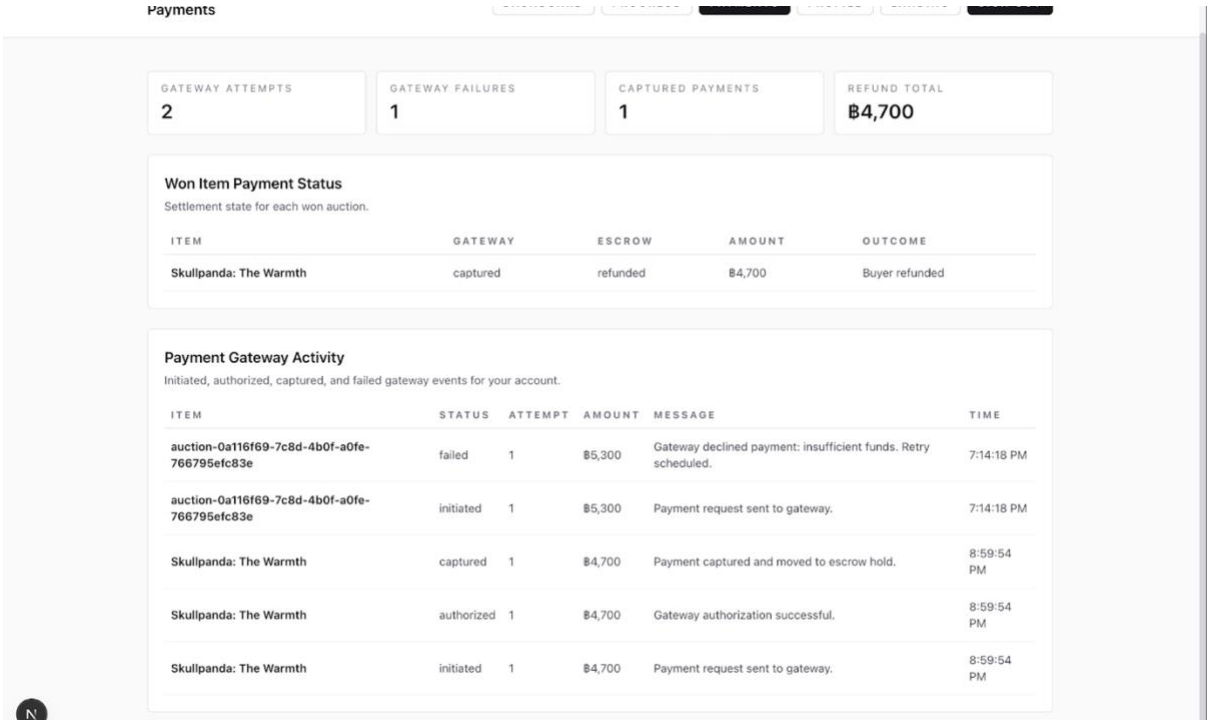

Figure 34: Fail payment process from buyers if they do not have enough fund for req_buy_9

11. Profile Management Page

Both buyers and sellers have a profile page to manage their data. Buyers can save shipping addresses and payment methods for faster checkout, while sellers can save payout details and pickup addresses for logistics and settlement

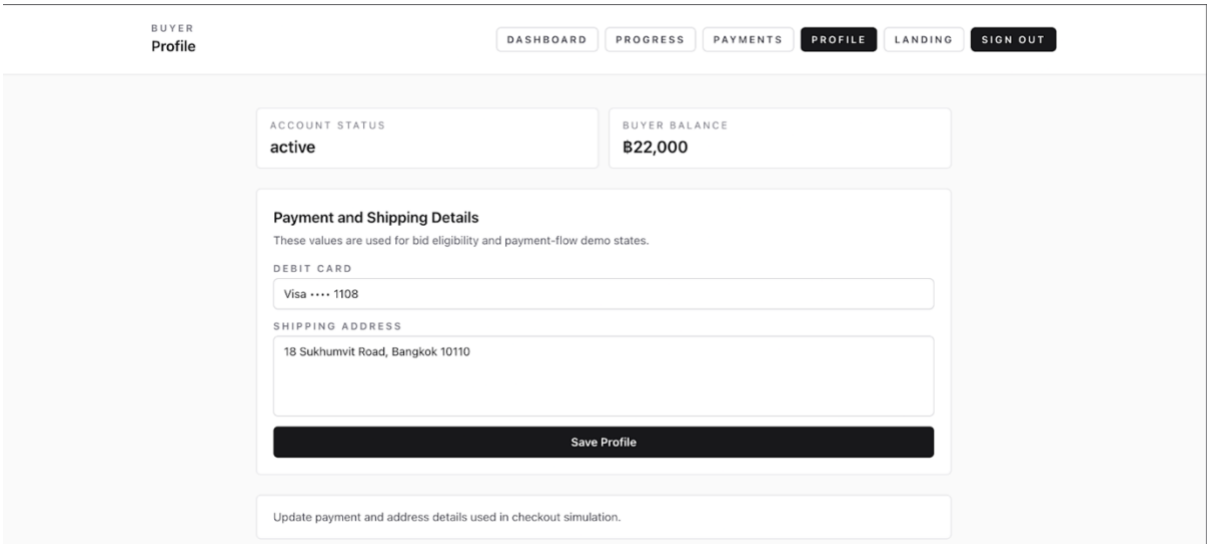

Figure 35: Payment Information and Address Management of Buyers for req_buy_8

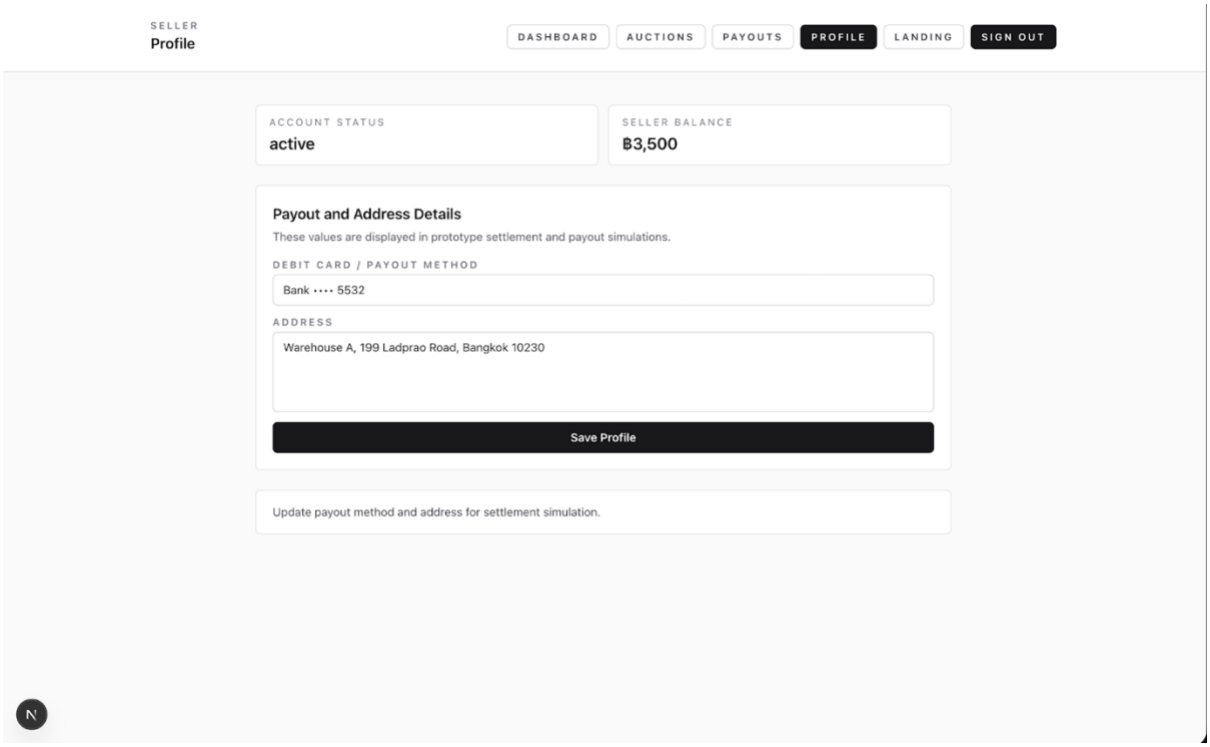

Figure 36: Pickup address and payment management of Sellers for req_sell_8

12. Seller Payment Tracking

After the auction has concluded and the buyer has processed the payment the seller can track the progress of the payment through this page.

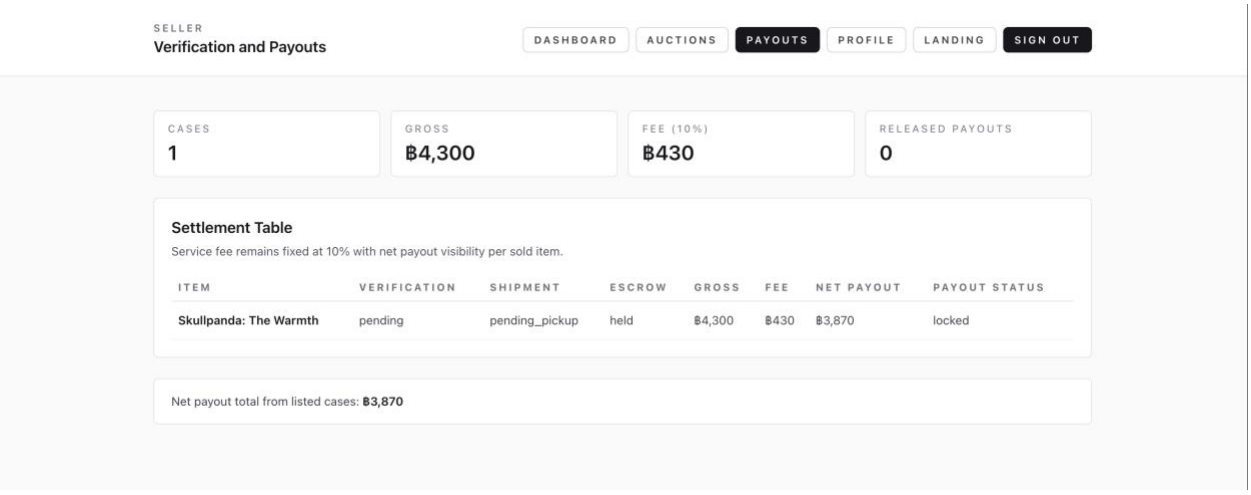

Figure 37: Seller payment tracking for req_seller_5

13. User Management

Admins can search for specific users and have the authority to suspend or ban accounts linked to fraudulent activity.

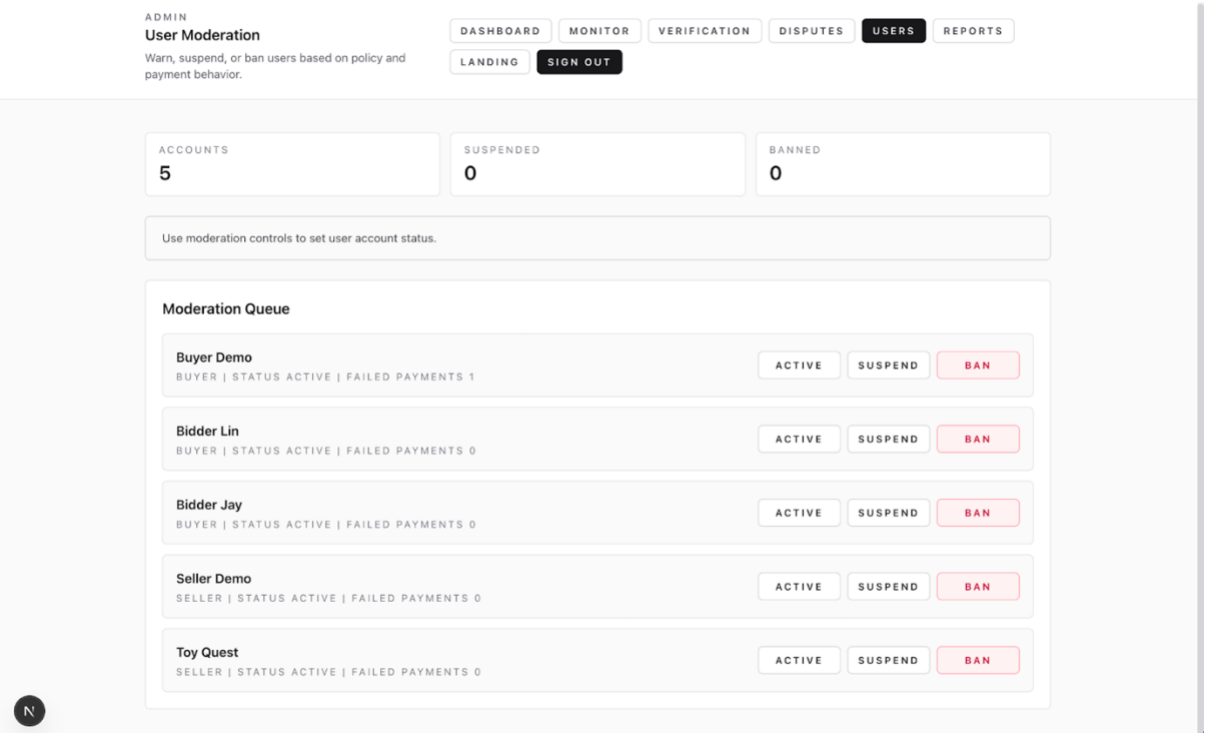

Figure 38: Banning and suspending users from admin workflow for req_admin_5

CONCLUSION

The Pop Mart Auction Platform is a digital solution for secure bidding, authenticity verification, and transparent trading of collectible pop mart figures. The system includes buyer, seller, and admin sections, each with specific functions. Buyers can participate in real-time auctions, receive outbid notifications, and track deliveries of verified items. Sellers can list their collectibles for auction, monitor verification status, and receive secure payouts through an escrow mechanism. Admin oversees the platform by screening listings, conducting physical authenticity checks, managing escrow payments, and resolving disputes. This system reduces fraud, ensures the authenticity of products, and enhances user confidence for both buyers and sellers in the collectible market.